Kernel Bayesian Robust Tensor Factorization for Hyperspectral Anomaly Detection

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***Abstract*—Hyperspectral anomaly detection (HAD) aims to identify targets whose spectral–spatial characteristics deviate from complex backgrounds and plays a crucial role in a wide range of remote sensing applications. While existing tensor-based HAD methods have shown strong performance, most rely on hand-crafted regularization to enforce local smoothness and require careful tuning of multiple model parameters, limiting their ability to characterize both global and local background correlations and increasing the burden of parameter tuning. To address these limitations, we propose a novel kernel Bayesian robust tensor factorization (KBRTF) method. Specifically, the background is modeled via a unitary transform-based low-tubal-rank tensor factorization to capture global spectral–spatial correlations, while Gaussian kernels are incorporated into the generation of the factor tensors, introducing distance-aware spatial correlations that encourage neighboring locations to share similar latent representations and thereby promote local spatial smoothness. Anomalies are represented by a sparse tensor with a hierarchical tube-wise prior that adaptively captures structured sparsity across spectral dimensions. Within the hierarchical Bayesian framework, most model parameters are adaptively learned during inference, thereby reducing the manual tuning burden. We develop an efficient variational Bayesian inference algorithm to estimate the posterior distributions of all latent variables. Extensive experiments on real hyperspectral datasets demonstrate that KBRTF achieves superior detection accuracy compared with state-of-the-art HAD methods.**

***Index Terms*—Hyperspectral anomaly detection (HAD), low-tubal-rank tensor, tensor factorization, variational Bayesian inference.**

NOMENCLATURE

$x, \boldsymbol{x}, \mathbf{X}, \mathcal{X}$	Scalar, vector, matrix, and tensor.
$\mathbb{R}, \mathbb{C}$	Sets of real and complex numbers.
$[K] = \{1, 2, \ldots, K\}$	Set of positive integer K.
$\mathcal{X}_{ijk}$	(i, j, k)th element of $\mathcal{X}$.
$\mathcal{X}_{:jk}, \mathcal{X}_{i:k}, \mathcal{X}_{ij:}$	Column, row, and tube fibers.
$\mathcal{X}_{i::}, \mathcal{X}_{:j:}, \mathcal{X}_{::k}$ (or $\mathbf{X}^{(k)}$)	Horizontal, lateral, and frontal slices.
$\mathcal{X}_{(n)}$ (or $\mathbf{X}_{(n)}$)	Mode-n unfolding of $\mathcal{X}$.
$\mathbf{I}_{n_3}$	$n_3 \times n_3$ identity matrix.
$\mathbf{1}_{n_1 \times n_2}$	$n_1 \times n_2$ all-ones matrix.
$\mathcal{Y} = \mathcal{X} \times_n \mathbf{A}$	Mode-n product (i.e., $\mathbf{Y}_{(n)} = \mathbf{A}\mathbf{X}_{(n)}$).
$\odot, \otimes$	Hadamard and Kronecker products.
$\|\mathcal{X}\|_F = \sqrt{\sum_{ijk} \|\mathcal{X}_{ijk}\|^2}$	Frobenius norm.
$\text{diag}(\boldsymbol{x})$	Diagonal matrix of $\boldsymbol{x}$.
$\text{tr}(\mathbf{X})$	Trace of $\mathbf{X}$.
$\text{vec}(\mathcal{X})$	Vectorization of $\mathcal{X}$.
$\mathbb{E}[x]$	Expectation of x.
$\mathcal{N}(y\|\mu, \sigma^2)$	Gaussian distribution.
$\text{Ga}(x\|a, b)$	Gamma distribution.

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I. INTRODUCTION

HYPERSPECTRAL images (HSIs) capture hundreds of spectral bands at each spatial location, enabling precise material identification and scene analysis in remote sensing. This rich spectral–spatial information has driven significant advances in classification [1], [2], [3], restoration [4], [5], [6], fusion [7], [8], [9], change detection [10], [11], [12], and anomaly detection [13], [14], [15], [16], [17], [18], [19], [20], [21]. Among these applications, hyperspectral anomaly detection (HAD) focuses on identifying pixels or regions whose spectral characteristics significantly deviate from their surrounding background, without requiring prior target knowledge or labeled samples. HAD has been widely studied in applications such as military surveillance [22], environmental monitoring [23], and camouflage recognition [24].

Early studies on HAD primarily relied on statistical background modeling to characterize normal spectral behavior and measure deviations from it. A seminal approach in HAD is the Reed-Xiaoli (RX) detector [25], which models the background using the multivariate Gaussian distribution and employs the Mahalanobis distance to assess the anomaly of pixels under test. On this basis, numerous RX variants have been proposed to enhance adaptability to diverse scenes, including local RX [26], dual-window RX [27], and multiwindow RX [28]. However, the inherent complexity and variability of real-world HSI data usually make it difficult to satisfy the Gaussian assumption. To address this issue, various extensions of RX have been introduced. Subspace RX [29] improved anomaly detection by projecting hyperspectral data into a

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lower-dimensional subspace, thereby separating background components and anomalies. The weighted-RX [30] enhanced background estimation by adjusting the weights of background samples, anomalous pixels, or noisy signals. The linear filter-based RX [30] suppressed false alarms by probabilistically filtering unreliable anomaly responses. In addition, the kernel RX (KRX) [31] incorporated a kernelization strategy that maps hyperspectral data into a high-dimensional feature space, enabling linear separation of anomalies in the transformed domain. Building upon this approach, the cluster KRX [32] further classified background pixels into multiple clusters using a fast eigendecomposition algorithm, which improves detection performance in spectrally heterogeneous scenes. In parallel, recent advances in deep learning have also stimulated the development of learning-based HAD methods, such as LREN [33], DeCNN-AD [34], PDBSNet [35], BS^{3}LNet [36], and GAED [37].

Additionally, extensive studies have explored the structural characteristics of HSIs to enhance anomaly detection. In real-world scenarios, backgrounds often exhibit strong spectral–spatial correlations, which can be effectively captured by low-rank models. In contrast, anomalies typically occupy only small regions and emerge as sparse deviations. Building on these properties, numerous structure-aware approaches based on low-rank and sparse decomposition have been proposed. A representative example is robust principal component analysis (RPCA) [38], which separates hyperspectral data into a low-rank background and a sparse anomaly component [39]. To improve robustness under complex noise conditions, LSMAD [40] introduced an additional noise term and leveraged the Mahalanobis distance for more effective anomaly discrimination. Later, LMSD-MOG [41] further advanced this idea by modeling noise as a mixture of Gaussian distributions, improving robustness in complex environments. In many practical scenarios, the hyperspectral background consists of multiple mixed sources, making the assumption of a single underlying subspace overly simplistic. To address this, LRASR [42] introduced a dictionary-based method capable of capturing both global and local structures. Similarly, GTVLRR [43] incorporated graph regularization and total variation (TV) to better preserve local geometry and spatial coherence. However, these methods require converting the 3-D HSI cube into a 2-D matrix, which inevitably leads to the loss of the inherent structure of HSIs.

As a natural extension of the matrix to higher-order space, tensor offers significant advantages in modeling multidimensional data [44], [45], [46], [47], [48], [49]. Unlike matrix-based approaches that flatten the data and lose interdimensional relationships, tensor formulation inherently preserves the spatial–spectral structure of the hyperspectral cube. Based on the tensor–tensor product (t-product) [50], [51], Wang et al. [52] introduced a tensor dictionary learning approach with adaptive low-rank constraints, enabling more flexible background modeling. On this basis, He et al. [53] introduced the ϵ-tensor nuclear norm (TNN), a tighter rank surrogate designed to more accurately recover low-rank components in hyperspectral scenes. Qin et al. [54] further improved representation quality by replacing the convex TNN with a generalized nonconvex surrogate, enhancing the expressiveness of the low-rank prior. While low-rank priors are effective for capturing the global structure in HSIs, real-world scenes often exhibit strong local continuity, stemming from gradual transitions in reflectance across neighboring regions. To better model this property, many methods incorporate local smoothness constraints, typically based on TV regularization. A representative example is the prior-based tensor approximation (PTA) model [55], which combines a TV norm for enforcing local smoothness, a truncated nuclear norm for modeling low-rank structure, and an $\ell_{2,1}$-norm to emphasize anomaly signatures. To jointly exploit the global low-rank structure and piecewise smoothness of the hyperspectral background, Sun et al. [56] proposed a tensor model that combines low average rank with TV regularization. Despite their effectiveness, existing tensor models typically rely on hand-crafted regularization and require careful tuning of multiple balance parameters, making them sensitive to noise and scene complexity. Moreover, most existing smoothness priors are formulated using linear neighborhood difference operators. For example, TV regularization encourages local smoothness by penalizing intensity differences between neighboring pixels. Although effective, such local difference operators primarily characterize piecewise smoothness and do not explicitly model the distance-aware spatial correlations commonly observed in hyperspectral backgrounds.

To overcome these limitations, this article proposes a kernel Bayesian robust tensor factorization (KBRTF) method for HAD. Motivated by the distinct physical characteristics of hyperspectral backgrounds and anomaly targets, our model decomposes the observed tensor into a low-tubal-rank background, a sparse anomaly component, and a noise term within a unified probabilistic framework. Since hyperspectral backgrounds typically exhibit strong global spectral–spatial correlations, the background tensor is modeled through a unitary transform-based low-tubal-rank tensor factorization. To further characterize local spatial continuity, Gaussian kernels are incorporated into the generation of the factor tensors. Specifically, the Gaussian kernel is defined as $\mathbf{K}_{i,j} = \exp(-((i-j)^2)/(\theta))$, where i and j index spatial positions (e.g., rows or columns) and θ is a spatial bandwidth parameter. This kernel assigns larger weights to nearby locations while gradually decreasing the influence of more distant locations, thereby introducing distance-aware spatial correlations and promoting local spatial smoothness in the reconstructed background. In contrast, anomaly targets generally occupy only a small fraction of the scene while exhibiting consistent spectral signatures across bands. To separate them effectively, we construct an explicit anomaly tensor and impose a hierarchical tube-wise sparsity-inducing prior, which captures the spectral consistency. Variational Bayesian inference is used for efficient posterior estimation over all latent variables. The main contributions of this work are summarized as follows.

1) We introduce a novel KBRTF method for HAD, which jointly captures low-tubal-rank background structure, distance-aware spatial correlations, and structured anomaly sparsity within a unified probabilistic framework.

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2) To characterize local spatial continuity in hyperspectral backgrounds, Gaussian kernels are incorporated into the tensor factorization process. By introducing distance-aware spatial correlations into the factor generation process, the proposed model promotes local spatial smoothness while preserving the global low-rank structure of the background.
3) We design a hierarchical tube-wise sparsity-inducing before govern the anomaly tensor, enforcing spectral coherency and enabling robust detection of structured anomalies under background interference and noise.
4) Variational Bayesian inference is employed to efficiently estimate posterior distributions and adaptively learn most model parameters during inference. In addition, the effective tubal rank is inferred through the hierarchical Bayesian framework. Extensive experiments conducted on several hyperspectral datasets validate the superiority of KBRTF compared with state-of-the-art methods.

The remainder of this article is structured as follows. We review related work in Section II, followed by an introduction to notations and preliminaries in Section III. Section IV describes the proposed KBRTF method, detailing the model formulation and inference procedure. Comprehensive experimental results on benchmark hyperspectral datasets and performance analysis are provided in Section V. Finally, Section VI concludes with a summary of key findings and potential directions for future work.

II. Related Work

Bayesian tensor factorization has attracted significant attention for modeling multidimensional data with uncertainty. Classical methods based on CP [57], [58], Tucker [59], [60], tensor train (TT) [61], and tensor ring (TR) [62], [63], [64] decompositions have demonstrated the ability to automatically infer rank, capture latent structures, and handle noisy or incomplete data. Based on the t-product, Zhou and Cheung [65] introduced a Bayesian framework for low-tubal-rank tensor modeling and demonstrated its effectiveness on real-world recovery tasks. In a separate line of work, Song et al. [66] highlighted the advantages of using unitary transforms to enhance the adaptability of low-tubal-rank modeling across a wider range of data types. Inspired by this, we consider a Bayesian formulation that incorporates unitary transforms to further improve the expressiveness and robustness of low-tubal-rank tensor factorization. This forms the basis of our proposed approach, which is specifically designed to meet the challenges of HAD.

Most existing Bayesian models have been developed for reconstruction tasks and do not consider the spatial continuity or structured variation inherent in hyperspectral data. Furthermore, their application to HAD is rare, hindering their effectiveness in scenarios where robust background modeling and structured anomaly separation are critical. Kernel-based Bayesian methods have emerged as powerful techniques for capturing nonlinear and locally coherent structures in multidimensional data. By projecting relationships into a high-dimensional feature space, kernels allow flexible modeling of spatial similarity and smooth transitions, which are particularly relevant for HSI. Gaussian kernels, Laplacian kernels, and graph-based kernels have been widely integrated into matrix and tensor models to enhance the description of spatial organization and preserve local continuity [67], [68], [69], [70], [71]. For HSI super-resolution, Ye et al. [72] introduced a Bayesian framework that utilizes a Gaussian process to characterize the relative smoothness of reflectance spectra. By defining a kernel based on wavelength differences, their method effectively handles spectral variability without requiring predefined degradation operators. Huang et al. [73] have incorporated kernel priors into a tensor decomposition framework and demonstrated its effectiveness in multidimensional data recovery tasks involving color images, face databases, and video sequences. Despite the effectiveness of these kernel-based Bayesian approaches, most existing methods are tailored to recovery or fusion tasks and do not address the specific demands of anomaly detection in HSIs. In such scenarios, anomalous regions often exhibit subtle deviations embedded within spatially coherent and spectrally redundant backgrounds. Capturing these deviations requires not only accurate global modeling but also flexible characterization of local spatial variation. To bridge this gap, we propose a kernel Bayesian tensor factorization model that jointly captures global low-tubal-rank structure and local spatial smoothness, enabling robust modeling for real-world hyperspectral scenes.

III. Notations and Preliminaries

We introduce the mathematical notations and foundational concepts used throughout this article. The notations for variables, sets, algebraic operators, and probability distributions are systematically summarized in the Nomenclature.

Let $\mathbf{U} \in \mathbb{C}^{n_3 \times n_3}$ be a unitary matrix, that is, $\mathbf{U}^H\mathbf{U} = \mathbf{U}\mathbf{U}^H = \mathbf{I}_{n_3}$, where $(\cdot)^H$ denotes the conjugate transpose. Applying this transform along the third dimension of a tensor $\mathcal{X} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$ maps it to the transformed domain. The resulting tensor $\mathcal{A}$ is computed by applying unitary matrix $\mathbf{U}$ to each tube fiber

$$\mathcal{A}_{ij:} = \mathbf{U}\mathcal{X}_{ij:} \quad \forall \quad i \in [n_1], \ j \in [n_2].$$

This operation can be expressed compactly as $\mathcal{A} = \mathbf{U}[\mathcal{X}]$. The inverse transform is then given by $\mathcal{X} = \mathbf{U}^H[\mathcal{A}]$. Within this transformed domain, several key operations are defined.

Definition 1 (Tensor Face-Wise Product): The tensor face-wise product [66] between two tensors $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_4}$ and $\mathcal{B} \in \mathbb{C}^{n_2 \times n_3 \times n_4}$ in the transformed domain, denoted by $\mathcal{A} \triangle \mathcal{B}$, results in a tensor $\mathcal{C} \in \mathbb{C}^{n_1 \times n_3 \times n_4}$. This product is equivalent to performing standard matrix multiplication on each pair of corresponding frontal slices

$$\mathbf{C}^{(k)} = \mathbf{A}^{(k)}\mathbf{B}^{(k)} \quad \forall k \in [K].$$

Definition 2 (Transformed Tensor Transpose): The conjugate transpose of a tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$ in the transformed domain, denoted $\mathcal{A}^H \in \mathbb{C}^{n_2 \times n_1 \times n_3}$, is obtained by taking the conjugate transpose of each of its frontal slices

$$(\mathcal{A}^H)^{(k)} = (\mathbf{A}^{(k)})^H \quad \forall k \in [K].$$

Definition 3 (Transformed Tubal Rank): The transformed tubal rank (or simply tubal rank) of a tensor $\mathcal{A} \in \mathbb{C}^{n_1 \times n_2 \times n_3}$,

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denoted $\text{rank}_t(\mathcal{A})$, is given by the maximum matrix rank across all of its frontal slices in the transformed domain [66]

$$\text{rank}_t(\mathcal{A}) = \max_{k\in[K]} \text{rank}(\mathbf{A}^{(k)}).$$

IV. Proposed Method

In this section, we present the hierarchical KBRTF framework for HAD. We first introduce the model specification, outlining the probabilistic generative process for the observed data. Subsequently, we derive the variational Bayesian inference algorithm for model learning and parameter estimation.

A. Model Specification

Let the observed HSI be denoted as a third-order tensor $\mathcal{Y} \in \mathbb{R}^{n_1\times n_2\times n_3}$, where n_1 and n_2 are the spatial dimensions and n_3 is the number of spectral bands. We model $\mathcal{Y}$ as the sum of a low-rank background component $\mathcal{L}$, a sparse anomaly component $\mathcal{S}$, and an additive noise term $\mathcal{E}$

$$\mathcal{Y} = \mathcal{L} + \mathcal{S} + \mathcal{E}. \tag{1}$$

The elements of the noise tensor $\mathcal{E}$ are assumed to follow an independent and identically distributed (i.i.d.) Gaussian distribution with precision parameter τ, that is, $\mathcal{E}_{ijk} \sim \mathcal{N}(0, \tau^{-1})$. The corresponding likelihood function for the observed data $\mathcal{Y}$ is

$$p(\mathcal{Y}|\mathcal{L}, \mathcal{S}, \tau) = \prod_{i}^{n_1}\prod_{j}^{n_2}\prod_{k}^{n_3} \mathcal{N}\left(\mathcal{Y}_{ijk}|\mathcal{L}_{ijk} + \mathcal{S}_{ijk}, \tau^{-1}\right). \tag{2}$$

A conjugate Gamma prior is assigned to the precision parameter τ

$$p(\tau) = \text{Ga}\left(\tau|a_0^\tau, b_0^\tau\right). \tag{3}$$

The background component $\mathcal{L}$, which exhibits strong spectral–spatial correlations, is modeled as a low-tubal-rank tensor. Accordingly, we factorize $\mathcal{L}$ in the transformed domain as the product of two smaller factor tensors $\mathcal{A} \in \mathbb{C}^{n_1\times r\times n_3}$ and $\mathcal{B} \in \mathbb{C}^{n_2\times r\times n_3}$

$$\mathcal{L} = \mathbf{U}^H\left[\mathcal{A} \triangle \mathcal{B}^H\right] \tag{4}$$

where $r \ll \min(n_1, n_2)$ denotes the tubal rank. To further refine the background model, we introduce a hierarchical Bayesian structure that incorporates spatial prior knowledge via Gaussian kernel matrices $\mathbf{K_G}$ and $\mathbf{K_H}$ along the row and column dimensions, respectively. The entries of these kernel matrices are determined by spatial distances between locations, such that nearby rows (or columns) receive similar kernel weights while the influence gradually decays with increasing distance. Consequently, neighboring spatial locations are coupled through similar weighted combinations of latent variables in the factor generation process, introducing distance-aware spatial correlations and promoting local spatial smoothness in the reconstructed background component. The factor tensors $\mathcal{A}$ and $\mathcal{B}$ are generated from latent tensors $\mathcal{G}$ and $\mathcal{H}$ as follows:

$$p\left(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g\right) = \prod_{i=1}^{n_1}\prod_{l=1}^{r}\prod_{k=1}^{n_3} \mathcal{N}\left(\mathcal{A}_{ilk}|\left(\mathbf{K_G}\right)_{i:}\mathcal{G}_{:lk},\ \sigma_g^{-1}\right) \tag{5}$$

$$p\left(\mathcal{B}|\mathcal{H}, \mathbf{K_H}, \sigma_h\right) = \prod_{j=1}^{n_2}\prod_{l=1}^{r}\prod_{k=1}^{n_3} \mathcal{N}\left(\mathcal{B}_{jlk}|\left(\mathbf{K_H}\right)_{j:}\mathcal{H}_{:lk},\ \sigma_h^{-1}\right). \tag{6}$$

We assign Jeffreys priors to the precision parameters σ_g and σ_h

$$p\left(\sigma_g\right) \propto \sigma_g^{-1} \quad \text{and} \quad p\left(\sigma_h\right) \propto \sigma_h^{-1}. \tag{7}$$

To adaptively reduce the effective tubal rank, we place automatic relevance determination (ARD) priors on the latent tensors $\mathcal{G}$ and $\mathcal{H}$

$$p(\mathcal{G}|\boldsymbol{\lambda}) = \prod_{i=1}^{n_1}\prod_{l=1}^{r} \mathcal{N}\left(\mathcal{G}_{il:}|\mathbf{0}, \left(\lambda_l\mathbf{I}_{n_3}\right)^{-1}\right) \tag{8}$$

$$p(\mathcal{H}|\boldsymbol{\lambda}) = \prod_{j=1}^{n_2}\prod_{l=1}^{r} \mathcal{N}\left(\mathcal{H}_{jl:}|\mathbf{0}, \left(\lambda_l\mathbf{I}_{n_3}\right)^{-1}\right) \tag{9}$$

where $\boldsymbol{\lambda} = [\lambda_1, \ldots, \lambda_r]$ and λ_l is the hyperparameter controlling the lth lateral slices of $\mathcal{G}$ and $\mathcal{H}$. Then, we place a conjugate Gamma prior on each λ_l independently

$$p(\boldsymbol{\lambda}) = \prod_{l=1}^{r} \text{Ga}\left(\lambda_l|a_0^\lambda, b_0^\lambda\right). \tag{10}$$

When λ_l takes a large value, the corresponding components in $\mathcal{G}$ and $\mathcal{H}$ are shrunk toward zero, leading to a reduction in the effective tubal rank.

Anomalies in HSIs are distinct targets with limited spatial area. To capture this structure, tube-wise sparsity is imposed on the anomaly tensor $\mathcal{S}$, treating each spectral tube $\mathcal{S}_{ij:}$ as a group. Each tube is assigned a Gaussian prior

$$p(\mathcal{S}|\boldsymbol{\xi}) = \prod_{i=1}^{n_1}\prod_{j=1}^{n_2} \mathcal{N}\left(\mathcal{S}_{ij:}\,\middle|\,\mathbf{0},\ \left(\xi_{ij}\mathbf{I}_{n_3}\right)^{-1}\right) \tag{11}$$

where $\boldsymbol{\xi}$ is the matrix of precision parameters. The conjugate Gamma priors are assigned to each ξ_{ij}

$$p(\boldsymbol{\xi}) = \prod_{i=1}^{n_1}\prod_{j=1}^{n_2} \text{Ga}\left(\xi_{ij}\middle|a_0^\xi, b_0^\xi\right). \tag{12}$$

This hierarchical specification encourages tube-wise sparsity across spectral bands. The precision parameters ξ_{ij} directly govern the prior variance of each spectral tube $\mathcal{S}_{ij:}$, thereby controlling its expected magnitude. As ξ_{ij} increases, the corresponding variance shrinks, exerting stronger shrinkage on $\mathcal{S}_{ij:}$ and pulling it closer to zero. This mechanism drives entire spectral tubes to vanish, thereby inducing structured sparsity.

Building on the likelihood function and hierarchical priors established above, the relationships among all variables in our framework are summarized by the probabilistic graphical model shown in Fig. 1. For notational simplicity, we group all latent variables and hyperparameters into the set $\boldsymbol{\Psi} = \{\mathcal{A}, \mathcal{B}, \mathcal{S}, \mathcal{G}, \mathcal{H}, \boldsymbol{\lambda}, \boldsymbol{\xi}, \tau, \sigma_g, \sigma_h\}$. The resulting joint distribution of the model can then be written as

$$\begin{aligned} p(\mathcal{Y}, \boldsymbol{\Psi}) = {} & p(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau)\ p\left(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g\right)\ p\left(\mathcal{B}|\mathcal{H}, \mathbf{K_H}, \sigma_h\right) \\ & \times p(\mathcal{S}|\boldsymbol{\xi})\ p(\mathcal{G}|\boldsymbol{\lambda})\ p(\mathcal{H}|\boldsymbol{\lambda}) \\ & \times p(\boldsymbol{\lambda})\ p(\boldsymbol{\xi})\ p(\tau)\ p\left(\sigma_g\right)\ p\left(\sigma_h\right). \end{aligned} \tag{13}$$

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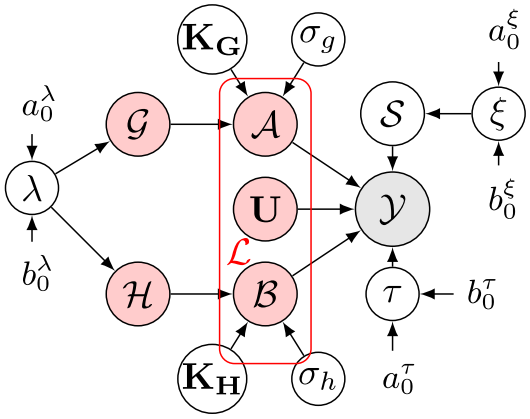


Fig. 1. Probabilistic graphical model of $\mathcal{Y}$.

B. Variational Inference for Model Learning

Exact posterior inference for this model is intractable. We therefore employ a variational Bayesian approach to approximate the posterior distribution $p(\mathbf{\Psi}|\mathcal{Y})$. Variational distribution $q(\mathbf{\Psi})$ is used to minimize the Kullback–Leibler (KL) divergence $\mathrm{KL}(q(\mathbf{\Psi})\|p(\mathbf{\Psi}|\mathcal{Y}))$. To enable tractable inference, we use a mean-field approximation, factorizing $q(\mathbf{\Psi}) = \prod_j q(\Psi_j)$. Then, the optimal form for each variable Ψ_j can be derived analytically by

$$\ln q\left(\Psi_j\right) = \mathbb{E}_{\mathbf{\Psi}\setminus\Psi_j}\left[\ln p\left(\mathcal{Y}, \mathbf{\Psi}\right)\right] + \text{const} \tag{14}$$

where $\mathbb{E}_{\mathbf{\Psi}\setminus\Psi_j}[\cdot]$ denotes the expectation over all variables except Ψ_j. This leads to an iterative algorithm where each variable is updated in turn. The detailed derivations for each posterior are provided in the Appendix. The major update rules are summarized below.

1) Posterior of Latent Tensors $\mathcal{G}$ and $\mathcal{H}$: For each column fiber of $\mathcal{G}$, the posterior $q(\mathcal{G}_{:lk})$ is a Gaussian distribution $\mathcal{N}(\mathcal{G}_{:lk}|\mathbb{E}[\mathcal{G}_{:lk}], \mathbf{\Sigma}^{\mathcal{G}_{:lk}})$, and the covariance and mean are represented as

$$\begin{aligned}\mathbf{\Sigma}^{\mathcal{G}_{:lk}} &= \left(\mathbb{E}\left[\sigma_g\right]\mathbf{K_G}^H\mathbf{K_G} + \mathbb{E}\left[\lambda_l\right]\mathbf{I}_{n_1}\right)^{-1}\\ \mathbb{E}\left[\mathcal{G}_{:lk}\right] &= \mathbb{E}\left[\sigma_g\right]\mathbf{\Sigma}^{\mathcal{G}_{:lk}}\mathbf{K_G}^H\mathbb{E}\left[\mathcal{A}_{:lk}\right].\end{aligned} \tag{15}$$

Similarly, for each column fiber of $\mathcal{H}$, the posterior $q(\mathcal{H}_{:lk})$ also follows a Gaussian distribution, and the covariance and mean are expressed as:

$$\begin{aligned}\mathbf{\Sigma}^{\mathcal{H}_{:lk}} &= \left(\mathbb{E}\left[\sigma_h\right]\mathbf{K_H}^H\mathbf{K_H} + \mathbb{E}\left[\lambda_l\right]\mathbf{I}_{n_2}\right)^{-1}\\ \mathbb{E}\left[\mathcal{H}_{:lk}\right] &= \mathbb{E}\left[\sigma_h\right]\mathbf{\Sigma}^{\mathcal{H}_{:lk}}\mathbf{K_H}^H\mathbb{E}\left[\mathcal{B}_{:lk}\right].\end{aligned} \tag{16}$$

2) Posterior of Factor Tensors $\mathcal{A}$ and $\mathcal{B}$: The posteriors of the factor tensors $\mathcal{A}$ and $\mathcal{B}$ are derived for their respective vectorized horizontal slices $\mathbf{a}_i$ and $\mathbf{b}_j$. To simplify the resulting expressions, we define the intermediate variables $T_j^{\mathcal{B}} = (\mathbf{U}^H \triangle \mathcal{B}_{j::})_{(1)}$ and $T_i^{\mathcal{A}} = (\mathbf{U}^H \triangle \mathcal{A}_{i::})_{(1)}$. In these operations, the unitary matrix $\mathbf{U}^H$ is treated as a tensor of size $n_3 \times 1 \times n_3$. The posterior of the ith vectorized slice $\mathbf{a}_i = \mathrm{vec}(\mathcal{A}_{i::})$ is a Gaussian distribution $q(\mathbf{a}_i) = \mathcal{N}(\mathbf{a}_i|\mathbb{E}[\mathbf{a}_i], \mathbf{\Sigma}^{\mathbf{a}_i})$, and the covariance and mean are denoted as

$$\begin{aligned}\mathbf{\Sigma}^{\mathbf{a}_i} &= \left(\mathbb{E}\left[\tau\right]\sum_{j=1}^{n_2}\mathbb{E}\left[\left(T_j^{\mathcal{B}}\right)^H T_j^{\mathcal{B}}\right] + \mathbb{E}\left[\sigma_g\right]\mathbf{I}_{rn_3}\right)^{-1}\\ \mathbb{E}\left[\mathbf{a}_i\right] &= \mathbf{\Sigma}^{\mathbf{a}_i}\left(\mathbb{E}\left[\tau\right]\sum_{j=1}^{n_2}\mathbb{E}\left[T_j^{\mathcal{B}}\right]^H\left(\mathcal{Y}_{ij:} - \mathbb{E}\left[\mathcal{S}_{ij:}\right]\right)\right.\\ &\qquad\left. + \mathbb{E}\left[\sigma_g\right](\mathbf{K_G})_{i:}\,\mathbb{E}\left[\mathcal{G}_{(1)}\right]\right)\end{aligned} \tag{17}$$

where $\mathbb{E}[(T_j^{\mathcal{B}})^H T_j^{\mathcal{B}}] = (\mathbb{E}[\mathcal{B}_{j::}]\mathbb{E}[\mathcal{B}_{j::}]^H + \Sigma^{\mathbf{b}_j}) \odot (\mathbf{I}_{n_3} \otimes \mathbf{1}_{r\times r})$.

Following a parallel derivation, the posterior of the slice $\mathbf{b}_j = \mathrm{vec}(\mathcal{B}_{j::})$ is also a Gaussian distribution $q(\mathbf{b}_j) = \mathcal{N}(\mathbf{b}_j|\mathbb{E}[\mathbf{b}_j], \mathbf{\Sigma}^{\mathbf{b}_j})$. Its covariance and mean are given as

$$\begin{aligned}\mathbf{\Sigma}^{\mathbf{b}_j} &= \left(\mathbb{E}\left[\tau\right]\sum_{i=1}^{n_1}\mathbb{E}\left[\left(T_i^{\mathcal{A}}\right)^H T_i^{\mathcal{A}}\right] + \mathbb{E}\left[\sigma_h\right]\mathbf{I}_{rn_3}\right)^{-1}\\ \mathbb{E}\left[\mathbf{b}_j\right] &= \mathbf{\Sigma}^{\mathbf{b}_j}\left(\mathbb{E}\left[\tau\right]\sum_{i=1}^{n_1}\mathbb{E}\left[T_i^{\mathcal{A}}\right]^H\left(\mathcal{Y}_{ij:} - \mathbb{E}\left[\mathcal{S}_{ij:}\right]\right)\right.\\ &\qquad\left. + \mathbb{E}\left[\sigma_h\right](\mathbf{K_H})_{j:}\,\mathbb{E}\left[\mathcal{H}_{(1)}\right]\right)\end{aligned} \tag{18}$$

where $\mathbb{E}[(T_i^{\mathcal{A}})^H T_i^{\mathcal{A}}] = (\mathbb{E}[\mathcal{A}_{i::}]\mathbb{E}[\mathcal{A}_{i::}]^H + \Sigma^{\mathbf{a}_i}) \odot (\mathbf{I}_{n_3} \otimes \mathbf{1}_{r\times r})$.

3) Posterior of Anomaly Tensor $\mathcal{S}$: The posterior $q(\mathcal{S}_{ij:})$ of each anomaly tube follows a Gaussian distribution $\mathcal{N}(\mathcal{S}_{ij:}|\mathbb{E}[\mathcal{S}_{ij:}], \mathbf{\Sigma}^{\mathcal{S}_{ij:}})$. Its covariance and mean are as follows:

$$\begin{aligned}\mathbf{\Sigma}^{\mathcal{S}_{ij:}} &= \left(\mathbb{E}\left[\tau\right] + \mathbb{E}\left[\xi_{ij}\right]\right)^{-1}\mathbf{I}_{n_3}\\ \mathbb{E}\left[\mathcal{S}_{ij:}\right] &= \mathbf{\Sigma}^{\mathcal{S}_{ij:}}\mathbb{E}\left[\tau\right]\left(\mathcal{Y}_{ij:} - \mathbf{U}^H\left(\mathbb{E}\left[\mathcal{A}_{i::}\right] \triangle \mathbb{E}\left[\mathcal{B}_{j::}^H\right]\right)\right).\end{aligned} \tag{19}$$

4) Posterior of Hyperparameters: Due to the conjugate prior structure of our model, the posterior distribution of each precision hyperparameter is a Gamma distribution. The updates for their respective parameters are detailed below.

The elements of $\boldsymbol{\lambda}$ are mutually independent, and the posterior for each component λ_l is Gamma distribution $\mathrm{Ga}(\lambda_l|a^{\lambda_l}, b^{\lambda_l})$, with its parameters computed as

$$\begin{aligned}a^{\lambda_l} &= a_0^\lambda + \frac{(n_1+n_2)\,n_3}{2}\\ b^{\lambda_l} &= b_0^\lambda + \frac{1}{2}\sum_{k=1}^{n_3}\left(\mathbb{E}\left[\|\mathcal{G}_{:lk}\|^2\right] + \mathbb{E}\left[\|\mathcal{H}_{:lk}\|^2\right]\right).\end{aligned} \tag{20}$$

The required expectations are computed as

$$\begin{aligned}\mathbb{E}\left[\|\mathcal{G}_{:lk}\|^2\right] &= \mathbb{E}\left[\mathcal{G}_{:lk}\right]^H\mathbb{E}\left[\mathcal{G}_{:lk}\right] + \mathrm{tr}\left(\mathbf{\Sigma}^{\mathcal{G}_{:lk}}\right)\\ \mathbb{E}\left[\|\mathcal{H}_{:lk}\|^2\right] &= \mathbb{E}\left[\mathcal{H}_{:lk}\right]^H\mathbb{E}\left[\mathcal{H}_{:lk}\right] + \mathrm{tr}\left(\mathbf{\Sigma}^{\mathcal{H}_{:lk}}\right).\end{aligned}$$

For the anomaly component, the posterior of each tube-wise precision $q(\xi_{ij})$ is a Gamma distribution $\mathrm{Ga}(\xi_{ij}|a^{\xi_{ij}}, b^{\xi_{ij}})$, with its parameters given as

$$a^{\xi_{ij}} = a_0^\xi + \frac{n_3}{2}, \quad b^{\xi_{ij}} = b_0^\xi + \frac{1}{2}\mathbb{E}\left[\left\|\mathcal{S}_{ij:}\right\|^2\right]. \tag{21}$$

The expectation term is calculated as

$$\mathbb{E}\left[\left\|\mathcal{S}_{ij:}\right\|^2\right] = \mathbb{E}\left[\mathcal{S}_{ij:}\right]^H\mathbb{E}\left[\mathcal{S}_{ij:}\right] + \mathrm{tr}\left(\mathbf{\Sigma}^{\mathcal{S}_{ij:}}\right).$$

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The posterior distributions for the kernel precision parameters σ_g and σ_h are also Gamma distributions. The posterior $q(\sigma_g) = \mathrm{Ga}(\sigma_g|a^{\sigma_g}, b^{\sigma_g})$ has the following parameters:

$$a^{\sigma_g} = \frac{n_1 r n_3}{2}, \quad b^{\sigma_g} = \frac{1}{2}\mathbb{E}\left[\|\mathcal{A} - \mathcal{G} \times_1 \mathbf{K_G}\|_F^2\right]. \tag{22}$$

The expectation in the update for b^{σ_g} is calculated as

$$\begin{aligned}\mathbb{E}\left[\|\mathcal{A} - \mathcal{G} \times_1 \mathbf{K_G}\|_F^2\right] &= \|\mathbb{E}[\mathcal{A}] - \mathbb{E}[\mathcal{G}] \times_1 \mathbf{K_G}\|_F^2 \\ &+ \sum_{i=1}^{n_1} \mathrm{tr}\left(\mathbf{\Sigma}^{\mathbf{a}_i}\right) \\ &+ \mathrm{tr}\left((\mathbf{K_G})^H \sum_{l=1}^{r}\sum_{k=1}^{n_3} \mathbf{\Sigma}^{\mathcal{G}_{:lk}} \mathbf{K_G}\right).\end{aligned}$$

Symmetrically, the posterior $q(\sigma_h) = \mathrm{Ga}(\sigma_h|a^{\sigma_h}, b^{\sigma_h})$ is given as

$$a^{\sigma_h} = \frac{n_2 r n_3}{2}, \quad b^{\sigma_h} = \frac{1}{2}\mathbb{E}\left[\|\mathcal{B} - \mathcal{H} \times_1 \mathbf{K_H}\|_F^2\right]. \tag{23}$$

The expectation in the update for b^{σ_h} is calculated as

$$\begin{aligned}\mathbb{E}\left[\|\mathcal{B} - \mathcal{H} \times_1 \mathbf{K_H}\|_F^2\right] &= \|\mathbb{E}[\mathcal{B}] - \mathbb{E}[\mathcal{H}] \times_1 \mathbf{K_H}\|_F^2 \\ &+ \sum_{j=1}^{n_2} \mathrm{tr}\left(\mathbf{\Sigma}^{\mathbf{b}_j}\right) \\ &+ \mathrm{tr}\left((\mathbf{K_H})^H \sum_{l=1}^{r}\sum_{k=1}^{n_3} \mathbf{\Sigma}^{\mathcal{H}_{:lk}} \mathbf{K_H}\right).\end{aligned}$$

For the noise precision τ, the posterior $q(\tau)$ is a Gamma distribution $\mathrm{Ga}(\tau|a^\tau, b^\tau)$, its parameters are updated as

$$\begin{aligned}a^\tau &= a_0^\tau + \frac{n_1 n_2 n_3}{2} \\ b^\tau &= b_0^\tau + \frac{1}{2}\sum_{i=1}^{n_1}\sum_{j=1}^{n_2}\mathbb{E}\left[\left\|\mathcal{Y}_{ij:} - \mathcal{S}_{ij:} - \mathbf{U}^H\left(\mathcal{A}_{i::} \triangle \mathcal{B}_{j::}^H\right)\right\|^2\right].\end{aligned} \tag{24}$$

The expectation of the squared reconstruction error is denoted as

$$\begin{aligned}&\mathbb{E}\left[\left\|\mathcal{Y}_{ij:} - \mathcal{S}_{ij:} - \mathbf{U}^H\left(\mathcal{A}_{i::} \triangle \mathcal{B}_{j::}^H\right)\right\|^2\right] \\ &= \left\|\mathcal{Y}_{ij:} - \mathbb{E}\left[\mathcal{S}_{ij:}\right] - \mathbf{U}^H\left(\mathbb{E}\left[\mathcal{A}_{i::}\right] \triangle \mathbb{E}\left[\mathcal{B}_{j::}^H\right]\right)\right\|^2 \\ &\quad - \left\|\mathbf{U}^H\left(\mathbb{E}\left[\mathcal{A}_{i::}\right] \triangle \mathbb{E}\left[\mathcal{B}_{j::}^H\right]\right)\right\|^2 + \mathrm{tr}\left(\mathbf{\Sigma}^{\mathcal{S}_{ij:}}\right) + \mathrm{tr}\left(\mathbb{E}\left[z_{ij} z_{ij}^H\right]\right)\end{aligned}$$

where $z_{ij} = \mathcal{A}_{i::} \triangle \mathcal{B}_{j::}^H$ and $\mathbb{E}[z_{ij} z_{ij}^H] = (\mathbf{I}_{n_3} \otimes \mathbf{1}_{1\times r})[(\mathbf{\Sigma}^{\mathbf{a}_i} + \mathbf{a}_i \mathbf{a}_i^H) \odot (\mathbf{\Sigma}^{\mathbf{b}_j} + \mathbf{b}_j \mathbf{b}_j^H)](\mathbf{I}_{n_3} \otimes \mathbf{1}_{r\times 1})$. The overall inference procedure is summarized in Algorithm 1.

Remark 1: The adaptive reduction of the effective tubal rank in Algorithm 1 is driven by the ARD prior. Specifically, λ_l serves as the precision (inverse variance) associated with the lth lateral slice of the latent tensors $\mathcal{G}$ and $\mathcal{H}$. During the variational Bayesian updates, if the lth component contributes little to reconstructing the background tensor, the posterior expectation of λ_l gradually increases to a large value. Consequently, the corresponding variance approaches zero, producing a strong shrinkage effect on the entire lateral slices $\mathcal{G}_{:l:}$ and $\mathcal{H}_{:l:}$. As these latent slices are suppressed, the corresponding components in the factor tensors $\mathcal{A}$ and $\mathcal{B}$ are also simultaneously eliminated. As a result, although the model is initialized with a specified rank upper bound r, redundant components are progressively removed during inference, yielding a substantially smaller effective tubal rank. An illustration of this rank-reduction mechanism is provided in Fig. 2. Furthermore, the evolution of $\log(\boldsymbol{\lambda})$ on the representative Airport-3 dataset [74] is visualized in Fig. 3, which clearly demonstrates the progressive elimination of redundant components throughout the iterative inference process.

Algorithm 1 KBRTF for HAD

Input: The observed HSI $\mathcal{Y}$.
Initialize $\mathcal{A}$, $\mathcal{B}$, $\mathcal{S}$, r, $\boldsymbol{\lambda}$, $\boldsymbol{\xi}$, τ, σ_g, and σ_h.
repeat
  Update posteriors $q(\mathcal{G})$ and $q(\mathcal{H})$ via (15) and (16).
  Update posterior $q(\lambda_l)$ via (20).
  Update posteriors $q(\mathcal{A})$ and $q(\mathcal{B})$ via (17) and (18).
  Update posterior $q(\sigma_g)$ and $q(\sigma_h)$ via (22) and (23).
  Update posterior $q(\mathcal{S})$ via (19).
  Update posterior $q(\xi_{ij})$ via (21).
  Update posterior $q(\tau)$ via (24).
  Adaptively reduce the tubal rank by monitoring the magnitude of entries in $\boldsymbol{\lambda}$.
until Termination criterion (25) is satisfied, or maximum iteration ($t = 100$) is reached.
Compute $\mathcal{L} = \mathbf{U}^H[\mathbb{E}[\mathcal{A}] \triangle \mathbb{E}[\mathcal{B}]^H]$ and $\mathcal{S} = \mathbb{E}[\mathcal{S}]$.
Output: Anomaly detection map $\mathbf{M}_{ij} = \sqrt{\sum_{k=1}^{n_3} \mathcal{S}_{ijk}^2}$.

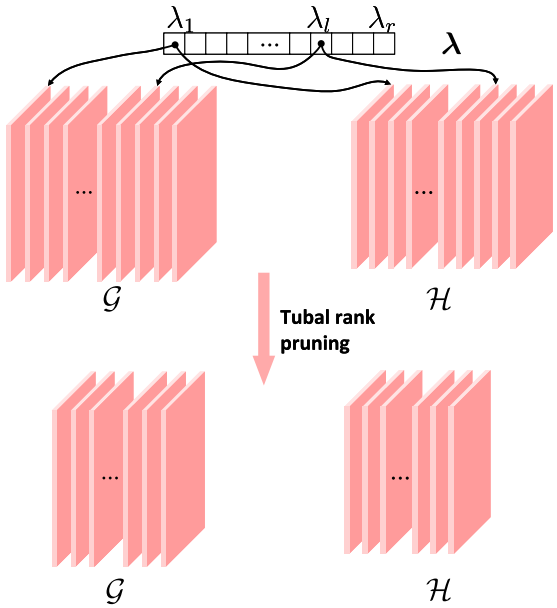


Fig. 2. Illustration of the effective tubal-rank reduction mechanism. Each hyperparameter λ_l controls the variance of the lth lateral slice in the latent tensors $\mathcal{G}$ and $\mathcal{H}$. During variational Bayesian inference, components that contribute little to background reconstruction are progressively shrunk toward zero, yielding a reduced effective tubal rank from the initialized rank upper bound.

V. Experimental Results

We evaluate the performance of the proposed KBRTF on eight benchmark HSI datasets in this section. All experiments

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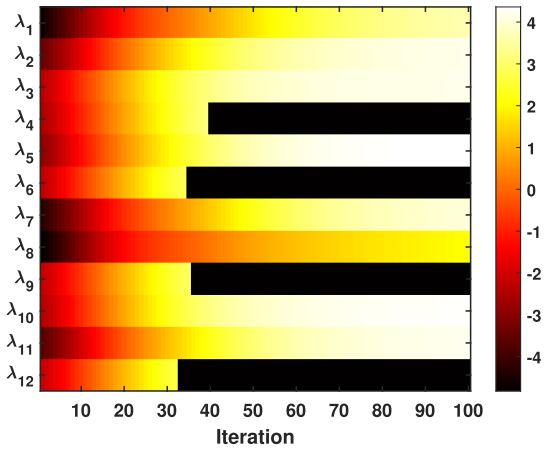


Fig. 3. Change of $\log(\lambda)$ values over 100 iterations. The bright bands represent active components that are kept, while the black blocks show exactly when unnecessary components are pruned and removed from the model.

are conducted using MATLAB R2023b on a workstation equipped with an Intel[1] Xeon[1] Gold-5122 3.60 GHz CPU and 192 GB RAM.

A. Datasets

All datasets employed in this study originate from real-world remote sensing scenarios. Before processing, each HSI is independently normalized across spectral bands. The false-color images and the corresponding ground-truth maps are shown in Fig. 4.

1) *Airport Dataset [74]:* This dataset was captured by the airborne visible/infrared imaging spectrometer (AVIRIS). Airplanes are labeled as anomalies, while the surrounding environment is considered background. The Airport-3 and Airport-4 have dimensions of 100 × 100 pixels with 205 and 191 spectral bands, respectively.
2) *Beach Dataset [74]:* The Beach-2 scene, collected by the AVIRIS sensor, features man-made targets such as ships, vehicles, and planes labeled as anomalies. Natural regions including beaches and water areas serve as background. It contains 193 bands at a spatial resolution of 100 × 100 pixels.
3) *Urban Dataset [74]:* The Urban-1, Urban-2, and Urban-3 scenes were captured by AVIRIS. Roofs are treated as anomalies. There are 204, 207, and 191 spectral bands for the three scenes, respectively, each with a spatial size of 100 × 100 pixels.
4) *AIR-HI Dataset*[2]: This scene was acquired using a visible and near-infrared imaging spectrometer, yielding 253 spectral bands over a full image of 3751 × 2746 pixels. Two subregions are selected: AIR-HI-Vehicle-I with a size of 80 × 100 and AIR-HI-Vehicle-II with a size of 80 × 120. Vehicles are labeled as anomalies, while grass, roads, and shadows are considered background.

[1]Registered trademark.
[2]https://github.com/MinghuaWang123/DSR-ADNR

TABLE I
RETAINED PCA DIMENSIONS FOR PCA-TLRSR, CF2-GNBRL, AND THE PROPOSED KBRTF ACROSS DIFFERENT DATASETS

Algorithms	Datasets							
	Airport-3	Airport-4	Beach-2	Urban-1	Urban-2	Urban-3	AIR-HI-Vehicle-I	AIR-HI-Vehicle-II
PCA-TLRSR	17	17	15	19	21	4	15	7
CF2-GNBRL	5	2	2	5	2	4	2	2
KBRTF	20	3	15	9	20	3	3	5

B. Detection Performance

To thoroughly validate the effectiveness of the proposed KBRTF method, we compare it with a range of representative HAD methods. The comparison includes one classic statistical detector (RX [25]), three matrix-based methods (LSMAD [40], LRASR [42], and GTVLRR [43]), three tensor-based methods (PTA [55], PCA-TLRSR [52], and CF2-GNBRL [18]), one deep learning-based method (GAED [37]), and one Bayesian method (LSDM-MoG [41]). For fairness, all baseline methods are implemented using their public code, with parameter configurations optimized based on guidelines provided in the original papers to ensure their best possible performance. For the proposed KBRTF method, the maximum iteration is set to 100. The termination criterion is defined as

$$\max\left\{\frac{\left\|\mathcal{L}^t-\mathcal{L}^{t-1}\right\|_F}{\left\|\mathcal{L}^{t-1}\right\|_F},\frac{\left\|\mathcal{S}^t-\mathcal{S}^{t-1}\right\|_F}{\left\|\mathcal{S}^{t-1}\right\|_F}\right\}<10^{-4} \tag{25}$$

where $\mathcal{L}^t$ and $\mathcal{S}^t$ represent the estimated low-rank and sparse components at the tth iteration, respectively. Hyperpriors are initialized as follows: $a_0^{\boldsymbol{\lambda}}=b_0^{\boldsymbol{\lambda}}=10^{-4}$, $a_0^{\boldsymbol{\xi}}=1$, $b_0^{\boldsymbol{\xi}}=10^{-5}$, and $a_0^{\tau}=10^{-4}, b_0^{\tau}=1$. The choice of parameters for KBRTF is based on empirical evaluation and is examined in detail in Section V-C. Moreover, to reduce computational complexity and improve detection efficiency, a principal component analysis (PCA)-based preprocessing strategy [52] is employed to project the input hyperspectral data onto a lower-dimensional subspace while preserving key spectral features. Specifically, among all the evaluated approaches, PCA-TLRSR, CF2-GNBRL, and our proposed KBRTF utilize this PCA technique. It is important to note that while these three methods share the same PCA preprocessing method, their retained principal component dimensions are not uniformly fixed. Instead, to ensure a rigorously fair comparison, the retained dimension for each algorithm is individually fine-tuned on each dataset to achieve its optimal detection performance. The specific retained PCA dimensions for these three methods across different datasets are detailed in Table I.

Performance evaluation is based on both qualitative and quantitative analyses, including anomaly detection maps, two types of receiver operating characteristic (ROC) curves, their corresponding area under the curve (AUC) values, and box-whisker plots. The first ROC curve is plotted between detection probability P_D and false alarm rate P_F, assessing the detector's sensitivity in identifying true anomalies. A curve that approximates the upper-left corner of the coordinate plane indicates higher detection capability. The second ROC curve is generated using the detection threshold η versus P_F, where lower-left positioning reflects stronger background suppression. AUC values are obtained by integrating the corresponding ROC curves along the horizontal axis, where

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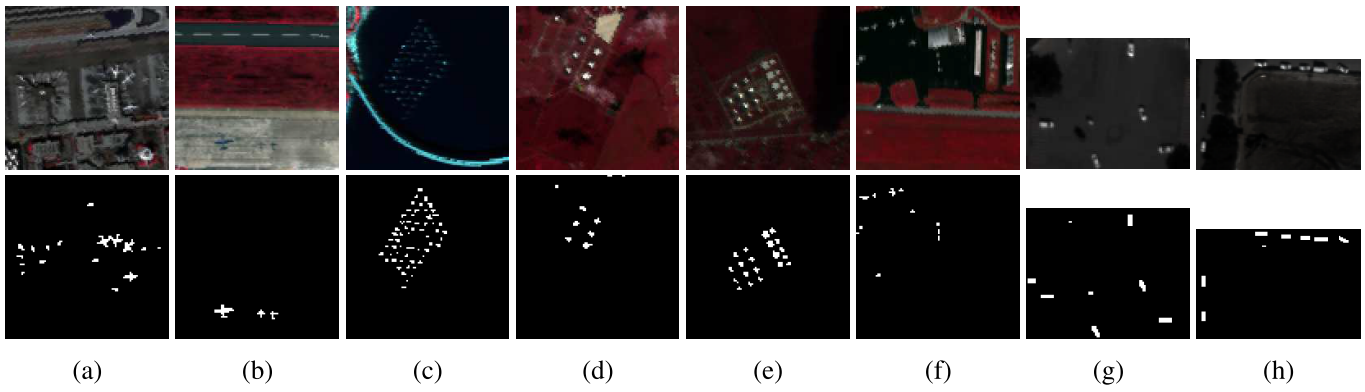


Fig. 4. False-color images and ground-truth maps of the eight HSI datasets. (a) Airport-3. (b) Airport-4. (c) Beach-2. (d) Urban-1. (e) Urban-2. (f) Urban-3. (g) AIR-HI-Vehicle-I. (h) AIR-HI-Vehicle-II.

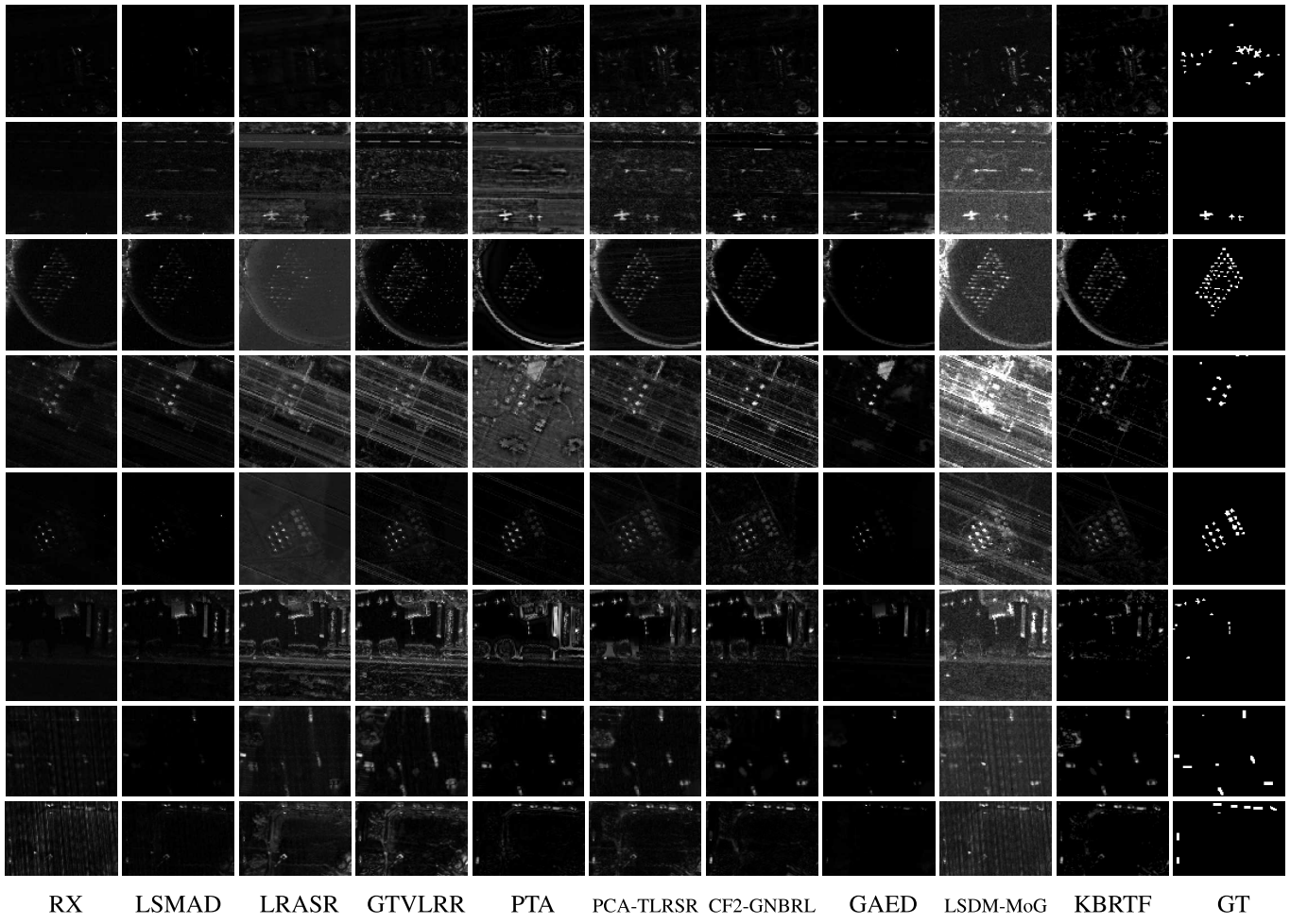


Fig. 5. Visual comparison of anomaly detection results obtained by different methods across eight HSI datasets. From top to bottom, each row corresponds to one test scene: Airport-3, Airport-4, Beach-2, Urban-1, Urban-2, Urban-3, AIR-HI-Vehicle-I, and AIR-HI-Vehicle-II. **Best viewed in** $\times 2$ **sized pdf file**.

higher values indicate better anomaly detection performance. Box-whisker plots are used to assess the separability between background and anomalous pixels. The boxes represent the tenth to 90th percentiles of the respective response distributions, with background shown in black and anomalies in red. Closer proximity of the narrow black box to the horizontal axis, together with greater separation from the red box, indicates stronger background suppression and more distinct anomaly discrimination. Based on the above evaluation metrics, a comprehensive summary of the detection performance is provided below.

Fig. 5 presents the visual comparison of anomaly detection results obtained by ten different methods across eight HSI datasets. By comparing these results with the ground truth, we can effectively evaluate the performance in terms of anomaly detection and background suppression. As shown in Fig. 5,

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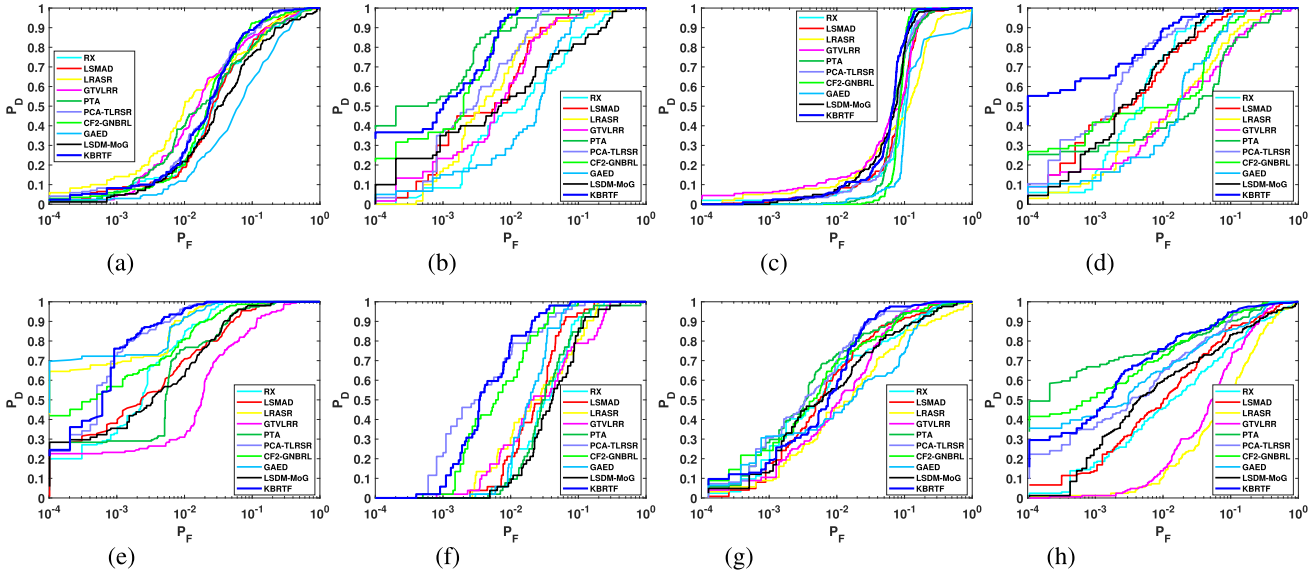


Fig. 6. ROC curves of (P_D, P_F) obtained by different comparison methods on eight datasets. (a) Airport-3. (b) Airport-4. (c) Beach-2. (d) Urban-1. (e) Urban-2. (f) Urban-3. (g) AIR-HI-Vehicle-I. (h) AIR-HI-Vehicle-II.

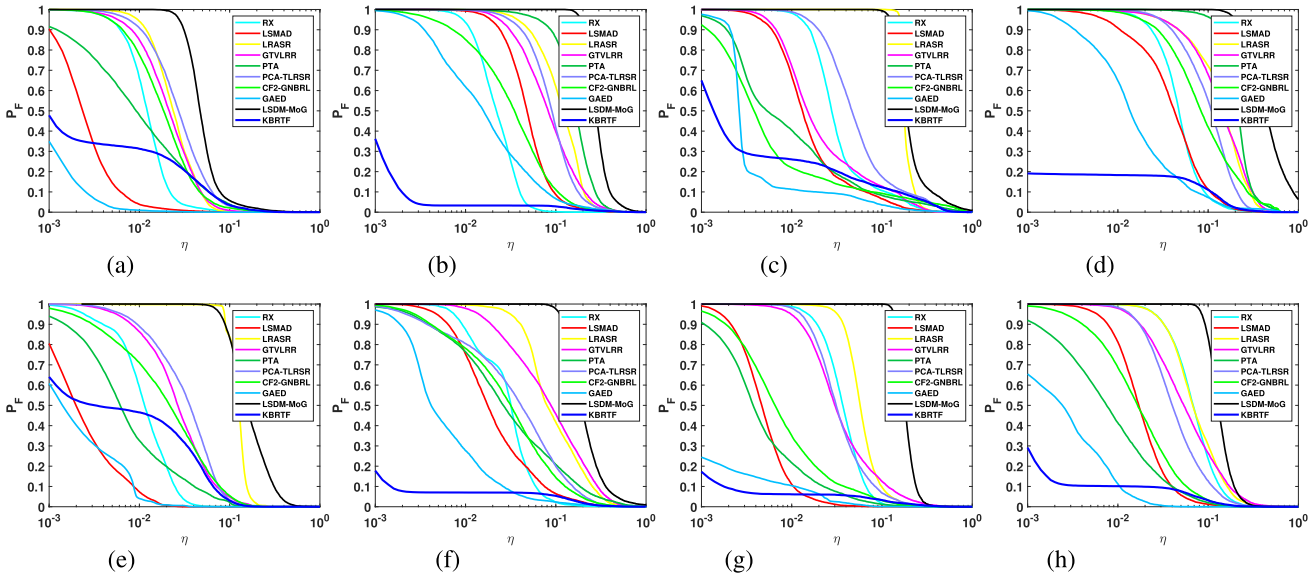


Fig. 7. ROC curves of (P_F, η) obtained by different comparison methods on eight datasets. (a) Airport-3. (b) Airport-4. (c) Beach-2. (d) Urban-1. (e) Urban-2. (f) Urban-3. (g) AIR-HI-Vehicle-I. (h) AIR-HI-Vehicle-II.

TABLE II

AUC VALUES OF (P_D, P_F) OBTAINED BY DIFFERENT COMPARISON METHODS ON EIGHT DATASETS

Datasets	RX	LSMAD	LRASR	GTVLRR	PTA	PCA-TLRSR	CF2-GNBRL	GAED	LSDM-MoG	KBRTF
Airport-3	0.9287	0.9327	0.9374	0.9380	0.9276	0.9540	0.9491	0.8791	0.8991	**0.9563**
Airport-4	0.9525	0.9874	0.9807	0.9839	0.9931	0.9931	0.9970	0.9681	0.9436	**0.9973**
Beach-2	0.9105	0.9047	0.8584	0.9177	0.9130	0.9148	0.9140	0.7868	0.9245	**0.9320**
Urban-1	0.9906	0.9837	0.9489	0.9322	0.9241	0.9942	0.9692	0.9696	0.9902	**0.9947**
Urban-2	0.9945	0.9832	0.9972	0.9562	0.9844	0.9982	0.9928	0.9971	0.9842	**0.9983**
Urban-3	0.9512	0.9650	0.9485	0.9329	0.9402	0.9849	0.9872	0.9751	0.9345	**0.9912**
AIR-HI-Vehicle-I	0.9585	0.9680	0.9138	0.9712	0.9683	**0.9852**	0.9704	0.9400	0.9519	0.9847
AIR-HI-Vehicle-II	0.9064	0.9399	0.8448	0.9061	0.9701	0.9682	0.9748	0.9565	0.9209	**0.9753**

the detection maps generated by the proposed KBRTF method are closer to the ground truth, whereas other methods exhibit varying degrees of limitations. For Airport-4, the RX method exhibits limited effectiveness, with anomalous targets being

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TABLE III
AUC VALUES OF (P_F, η) OBTAINED BY DIFFERENT COMPARISON METHODS ON EIGHT DATASETS

Datasets	RX	LSMAD	LRASR	GTVLRR	PTA	PCA-TLRSR	CF2-GNBRL	GAED	LSDM-MoG	KBRTF
Airport-3	0.0144	0.0037	0.0274	0.0261	0.0222	0.0349	0.0253	**0.0013**	0.0568	0.0184
Airport-4	0.0247	0.0577	0.1286	0.1057	0.1759	0.1048	0.0507	0.0346	0.2983	**0.0083**
Beach-2	0.0488	0.0290	0.1878	0.0432	0.0404	0.0811	0.0450	**0.0144**	0.2298	0.0431
Urban-1	0.0555	0.0538	0.1615	0.1635	0.2264	0.1187	0.1259	0.0348	0.5138	**0.0241**
Urban-2	0.0135	0.0036	0.1266	0.0356	0.0153	0.0405	0.0326	**0.0033**	0.1812	0.0234
Urban-3	0.0350	0.0353	0.1208	0.1270	0.0712	0.0644	0.0546	0.0143	0.2350	**0.0127**
AIR-HI-Vehicle-I	0.0410	0.0063	0.0658	0.0525	0.0107	0.0439	0.0172	**0.0036**	0.1877	0.0092
AIR-HI-Vehicle-II	0.0785	0.0222	0.0869	0.0797	0.0162	0.0548	0.0262	**0.0043**	0.1390	0.0097

TABLE IV
RUNNING TIME (S) OF DIFFERENT COMPARISON METHODS ON EIGHT DATASETS

Datasets	RX	LSMAD	LRASR	GTVLRR	PTA	PCA-TLRSR	CF2-GNBRL	GAED	LSDM-MoG	KBRTF
Airport-3	**0.12**	15.60	45.02	219.87	50.11	14.49	131.57	66.07	12.18	50.59
Airport-4	**0.28**	13.95	48.83	233.25	40.71	14.94	109.60	61.34	12.57	7.70
Beach-2	**0.10**	14.26	48.99	231.25	42.19	12.71	92.59	93.05	21.37	41.61
Urban-1	**0.19**	15.12	48.87	278.92	42.38	15.90	121.88	30.23	24.59	21.41
Urban-2	**0.30**	15.06	45.20	192.79	45.86	17.68	108.48	99.64	25.60	57.55
Urban-3	**0.23**	14.25	43.63	246.30	41.12	3.62	116.83	61.82	16.80	8.96
AIR-HI-Vehicle-I	**0.46**	13.87	40.30	152.55	49.32	12.33	108.85	58.76	9.78	6.98
AIR-HI-Vehicle-II	**0.12**	16.83	53.95	237.16	58.80	6.66	157.68	67.57	13.24	12.51

barely visible. Similarly, the LSMAD and GAED methods fail to accurately capture the complete outlines of the aircraft. Although the LRASR, GTVLRR, PTA, PCA-TLRSR, CF2-GNBRL, and LSDM-MoG methods provide clear aircraft profiles, their detection maps contain a large number of white noise spots. These areas correspond to background pixels that are incorrectly classified as anomalies. For Urban-1, the PTA detector fails to effectively identify the anomalous targets. The detection maps generated by LRASR, GTVLRR, PCA-TLRSR, CF2-GNBRL, and LSDM-MoG exhibit pronounced stripe artifacts, primarily due to background features being incorrectly identified as anomalies. In contrast, RX and LSMAD maintain a relatively clean background, but the visual intensity of the anomalous targets is weak. Both GAED and KBRTF demonstrate detection performance that is significantly superior to the other comparative methods. However, a careful inspection reveals that GAED tends to misclassify certain background pixels as anomalies, particularly in the upper regions of the scene. In contrast, the proposed KBRTF method effectively avoids such false detections. Overall, KBRTF accurately identifies distinct anomalous targets while effectively suppressing background interference.

Fig. 6 shows that the proposed KBRTF method achieves higher detection rates than competing methods across varying P_F values in most cases, with ROC curves of (P_D, P_F) approaching the upper-left corner. Fig. 7 further demonstrates that the ROC curves of (P_F, η) are positioned closer to the lower-left corner, highlighting the effectiveness of KBRTF in suppressing background responses. The AUC values of (P_D, P_F) and (P_F, η) are summarized in Tables II and III, respectively, with optimal and suboptimal results marked in bold and underlined. KBRTF achieves the best or second-best AUC values of (P_D, P_F) across all eight benchmark datasets, demonstrating superior detection accuracy. Furthermore, KBRTF attains competitive AUC values of (P_F, η), confirming its capability to detect anomalies while effectively suppressing background pixels. The computational efficiency of different methods is compared in Table IV by reporting their running time on eight datasets. As can be observed, RX consistently achieves the shortest runtime across all datasets. In contrast, several methods exhibit substantially higher computational cost, with GTVLRR and CF2-GNBRL being the most time-consuming. Notably, the proposed KBRTF method demonstrates competitive efficiency on most datasets. In Airport-4 and AIR-HI-Vehicle-I, KBRTF is one of the fastest methods besides RX, which requires less computation time. These results indicate that KBRTF achieves strong detection capability while maintaining high computational efficiency.

To further quantify the discrimination capability of different methods, box-whisker maps are presented in Fig. 8. The proposed KBRTF method achieves more distinct separation between background (black) and anomaly (red) response boxes compared to other detectors across most scenarios. For Airport-3, although all methods exhibit vanishing interbox

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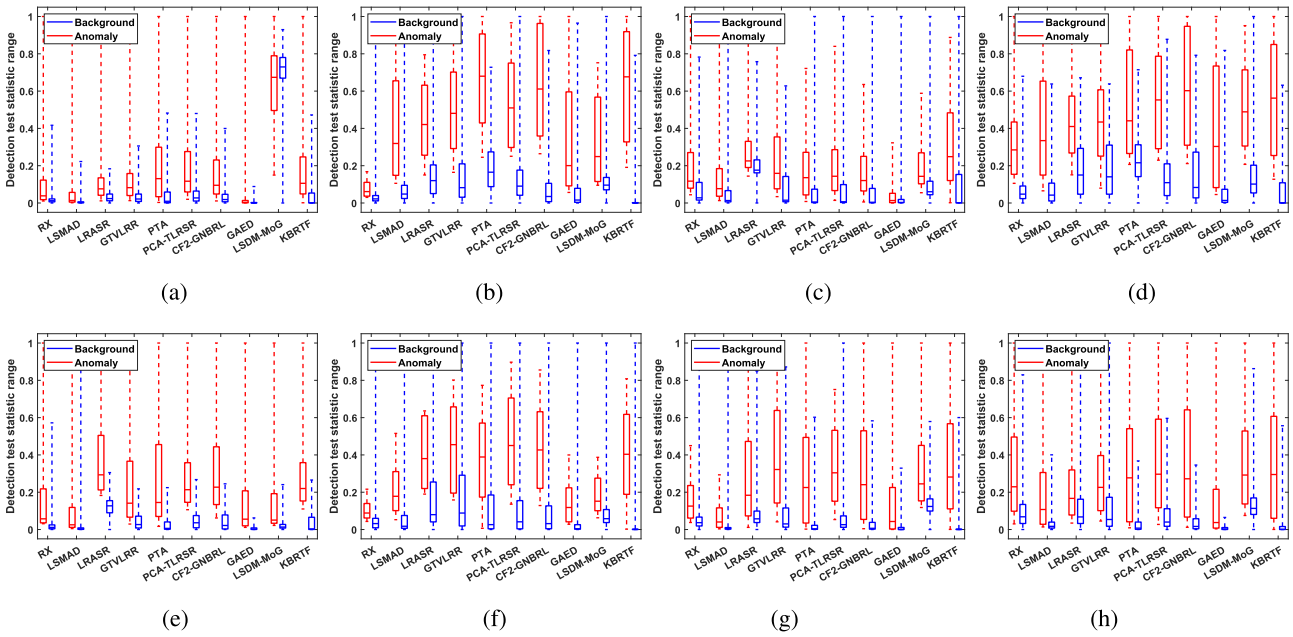


Fig. 8. Box-whisker maps for different comparison methods on eight datasets. (a) Airport-3. (b) Airport-4. (c) Beach-2. (d) Urban-1. (e) Urban-2. (f) Urban-3. (g) AIR-HI-Vehicle-I. (h) AIR-HI-Vehicle-II.

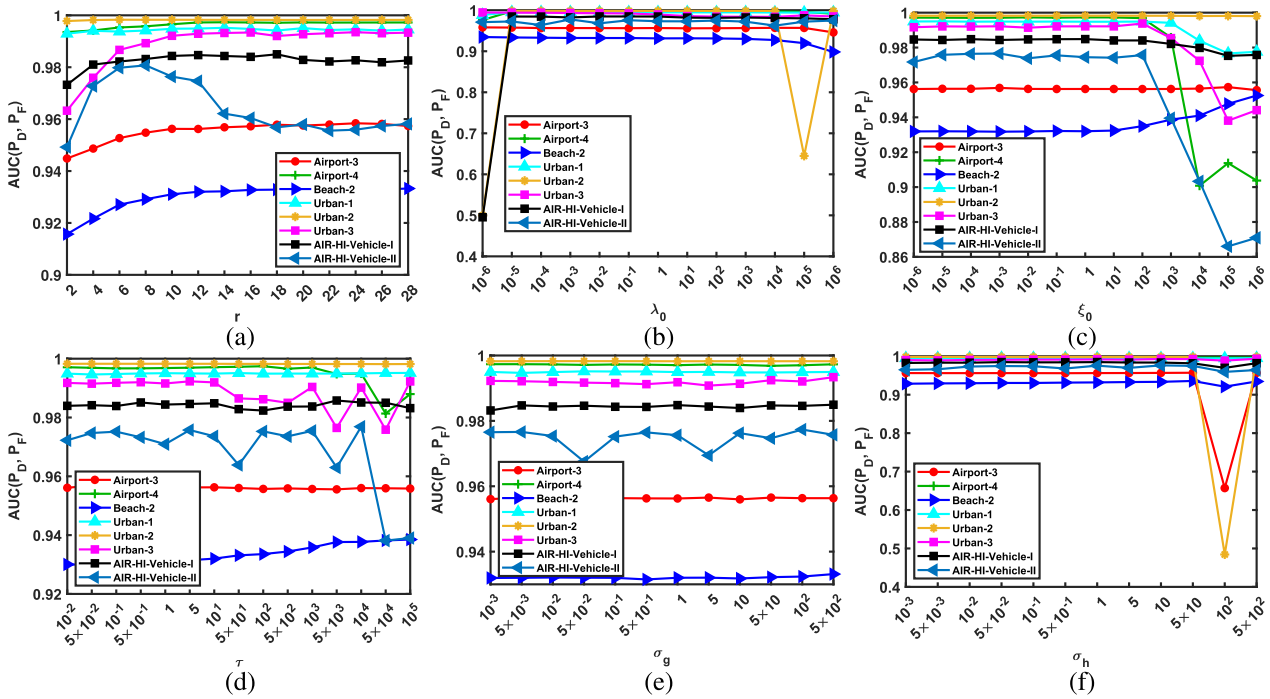


Fig. 9. Sensitivity analysis of the proposed method with respect to different parameters. (a) r. (b) λ_0. (c) ξ_0. (d) τ. (e) σ_g. (f) σ_h.

gaps, KBRTF demonstrates minimal overlap between distributions. For Beach-2, the anomaly box for KBRTF is positioned notably higher than those of competing methods, indicating enhanced responsiveness to anomalous pixels. Notably, for Airport-4, Urban-3, and AIR-HI-Vehicle-I, the background boxes are nearly aligned with the horizontal axis, reflecting superior background suppression capability. These quantitative observations confirm the previous visual analysis and further verify the effectiveness of the proposed KBRTF method.

C. Parameter Analysis

The performance of the proposed method depends on the initialization of several key parameters: the initial tubal rank r, and the initial prior parameters τ, σ_g, and σ_h. The initial parameter vectors $\boldsymbol{\lambda}$ and $\boldsymbol{\xi}$ are initialized using $\boldsymbol{\lambda} = \lambda_0 \times \text{rand}(r, 1)$ and $\boldsymbol{\xi} = \xi_0 \times \text{rand}(n_1, n_2)$, respectively. It is worth noting that while these parameters are set initially, they are dynamically updated and inferred in our hierarchical Bayesian framework during the variational Bayesian iterations.

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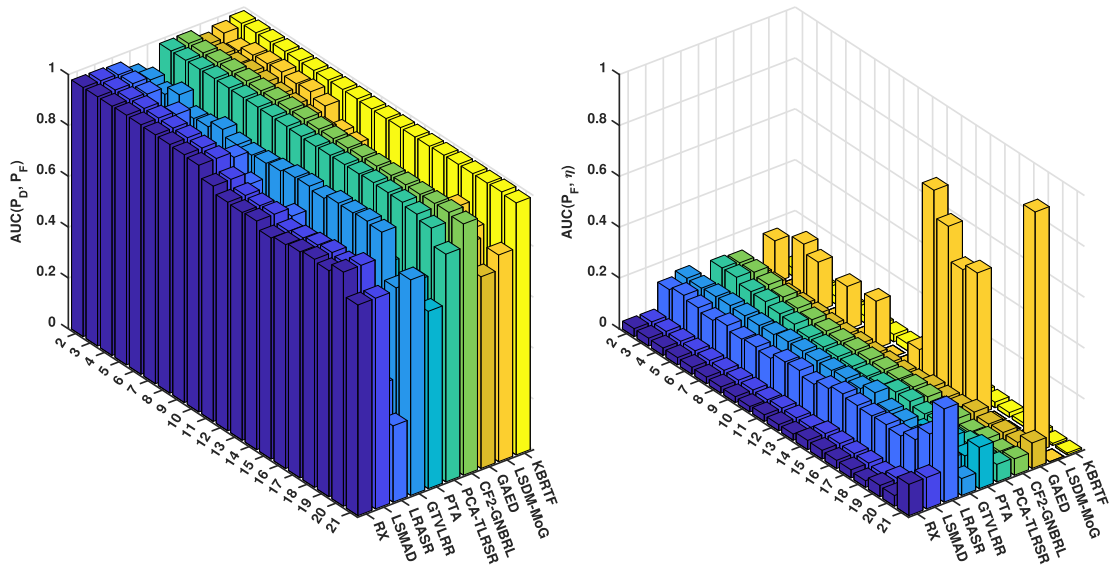


Fig. 10. AUC values of all comparison methods when uniformly applying the same PCA preprocessing with varying retained dimensions (ranging from 2 to 21).

TABLE V
FINAL EFFECTIVE TUBAL RANK AFTER PRUNING ACROSS THE EIGHT DATASETS (WITH INITIAL RANK $r = 12$)

Datasets	Airport-3	Airport-4	Beach-2	Urban-1	Urban-2	Urban-3	AIR-HI-Vehicle-I	AIR-HI-Vehicle-II
Effective tubal rank	8	12	12	11	12	11	4	7

To analyze the algorithm's sensitivity to its initial parameter settings, we conduct a series of experiments on eight hyperspectral datasets. In this analysis, each parameter is varied individually within a predefined range, while the remaining parameters are fixed at their optimal values. The results are presented in Fig. 9, where the X-axis denotes the parameter values and the Y-axis represents AUC(P_D, P_F).

To assess the influence of the initial tubal rank r on detection performance, we vary its value from 2 to 28 with a step of 2, as depicted in Fig. 9(a). We can see that the AIR-HI-Vehicle-II shows considerable performance variation as r changes. Based on a comprehensive analysis of the results, we identify $r = 12$ as a suitable initial choice. While KBRTF can adaptively determine the rank starting from a large value like $r = \min(n_1, n_2)$ (as analyzed in Section V-E5), setting a moderate initial rank $r = 12$ significantly reduces the computational complexity of the first few iterations. The initial rank of 12 is still adaptively pruned down to the optimal effective rank for each dataset, as reported in Table V. The empirical evolution of the precision parameters $\log(\boldsymbol{\lambda})$ over iterations on the representative Airport-3 dataset has been illustrated in Fig. 3 (with initial $r = 12$), and its conceptual pruning mechanism is shown in Fig. 2. These figures demonstrate how the automatic rank determination mechanism effectively prunes the massive initial rank down to a very small rank, keeping the r-related computations highly efficient.

The algorithm's sensitivity to the initial settings of the parameters $\lambda_0, \xi_0, \tau, \sigma_g$, and σ_h is detailed in Fig. 9(b)–(f). Fig. 9(b) and (c) demonstrates this stability clearly. The AUC(P_D, P_F) values are consistently high for initial λ_0 ranging from 10^{-5} to 10^4 and for initial ξ_0 ranging from 10^{-6} to 10^2. Beyond these ranges, performance degradation becomes apparent, particularly for ξ_0, where several datasets exhibit a sharp decline after the 10^2 threshold. Fig. 9(d) illustrates the effect of the parameter τ on the detection performance, with its value tested across a wide range from 10^{-2} to 10^5. A key observation is that the algorithm's performance is largely insensitive to the choice of initial τ for a majority of the datasets. For instance, the curves for Airport-3, Urban-1, Urban-2, and AIR-HI-Vehicle-I remain almost flat, indicating high stability. The Beach-2 exhibits a slight but consistent improvement in AUC(P_D, P_F) as τ increases. In contrast, some datasets, such as Urban-3 and AIR-HI-Vehicle-II, exhibit notable fluctuations, particularly at larger values of τ. Finally, Fig. 9(e) and (f) shows the impact of the initial parameters σ_g and σ_h, respectively, which are tested in the range of $\{10^{-3}, 5 \times 10^{-3}, \ldots, 5 \times 10^2\}$. Overall, the algorithm is not particularly sensitive to these parameters. As seen in Fig. 9(e), the performance remains highly stable for the initial σ_g across the entire test range. A similar stability is observed for the initial σ_h in Fig. 9(f), with the only exception being a sharp performance drop when its value becomes very large (e.g., 10^2). Based on the comprehensive parameter sensitivity analysis across all tested ranges, the proposed algorithm is highly insensitive to these initial settings over a wide range. Consequently, we set the initial prior parameters as $\lambda_0 = 0.1$, $\xi_0 = 1$, $\tau = 10$, and $\sigma_g = \sigma_h = 1$ in all our experiments.

D. PCA Preprocessing Analysis

To validate the necessity of the dimensionality reduction step, we evaluate our proposed KBRTF directly on the full-band Airport-4 dataset. As presented in Table VI, applying PCA significantly decreases the computational time due to the reduced data volume in the spectral domain. Moreover, the detection performance is also improved when using PCA. This improvement implies that projecting the data into a principal

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TABLE VI
DETECTION PERFORMANCE AND RUNNING TIME (S) OF THE PROPOSED METHOD WITH AND WITHOUT PCA ON EIGHT DATASETS

Datasets	AUC (P_D, P_F)		Running Time (s)	
	Without PCA	With PCA	Without PCA	With PCA
Airport-3	0.9026	**0.9563**	447.0749	**50.5871**
Airport-4	0.9907	**0.9973**	402.3089	**7.7032**
Beach-2	0.9266	**0.9320**	398.9466	**41.6143**
Urban-1	0.9825	**0.9947**	439.0791	**21.4058**
Urban-2	0.9856	**0.9983**	441.2468	**57.5516**
Urban-3	0.9576	**0.9912**	392.7715	**8.9591**
AIR-HI-Vehicle-I	0.9568	**0.9847**	565.2675	**6.9812**
AIR-HI-Vehicle-II	0.8236	**0.9753**	603.5492	**12.5130**

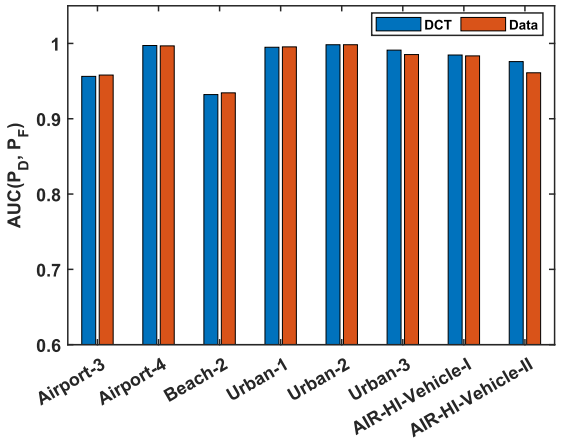


Fig. 11. Detection performance of the proposed method with different unitary transform strategies on eight datasets.

subspace effectively filters out redundant spectral interference, facilitating a more robust tensor factorization for anomaly detection. Further, we apply the same PCA to all compared methods on the Airport-4 dataset. Specifically, we evaluate the performance of all methods with the retained dimension varying from 2 to 21, and report both the AUC(P_D, P_F) and AUC(P_F, η). The left side of Fig. 10 shows that CF2-GNBRL and KBRTF maintain stable and relatively high AUC(P_D, P_F) across varying dimensions. The right side of Fig. 10 presents the AUC(P_F, η) results. When the retained dimension is 2, both LSDM-MoG and KBRTF yield low values, while as the dimension increases, LSDM-MoG undergoes a sharp jump in AUC(P_F, η). Overall, under the same PCA settings, our KBRTF method shows good detection performance, proving that the PCA preprocessing is very useful for our method.

E. Discussions

1) Different Unitary Transform Representation: We investigate the impact of different unitary transform representations on HAD performance by comparing a fixed analytical transform and a data-adaptive alternative. Specifically, the discrete cosine transform (DCT) is evaluated against a data-driven unitary transform, which is constructed by learning an orthogonal basis from the observed tensor data following the procedure in [75]. As shown in Fig. 11, the proposed KBRTF method exhibits consistently high AUC(P_D, P_F) values under both transform choices across all datasets, indicating that the model is not overly sensitive to the specific form of the unitary transform. In most scenarios, the DCT-based implementation achieves slightly superior or comparable detection accuracy relative to its data-driven counterpart. Accordingly, the DCT is adopted as the default transform in our experiments.

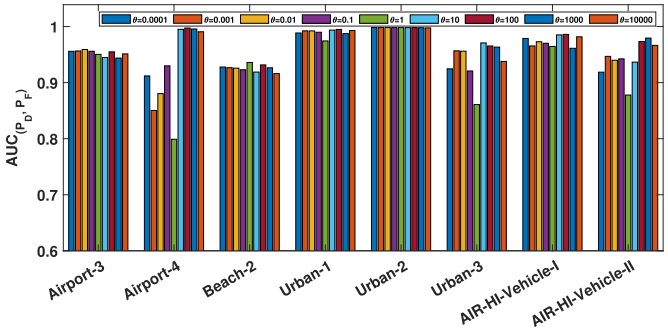


Fig. 12. Detection performance of the proposed method with different kernel parameter θ on eight datasets.

2) Kernel Parameter Analysis: In the proposed Bayesian method, τ, σ_g, σ_h, $\boldsymbol{\lambda}$, and $\boldsymbol{\xi}$ only require initialization and are subsequently inferred through iterative variational Bayesian updates. In contrast, the kernel parameter θ is manually specified and remains fixed during inference. To evaluate its impact and assess the sensitivity of the method to different kernel bandwidth values, we evaluate the detection performance over a logarithmically spaced range $\theta \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 10^2, 10^3, 10^4\}$. The evaluation results across all eight datasets are shown in Fig. 12. For Airport-4, setting a very small bandwidth ($\theta \leq 1$) leads to a noticeable performance drop (with AUC falling to $0.80 \sim 0.93$), because an under-smoothed kernel fails to capture the local spatial smoothness of continuous runways and grasslands. For AIR-HI-Vehicle-II, we observe a slight performance dip at $\theta = 1$, but it maintains highly stable and excellent performance at all other tested values. Based on these empirical observations, we select $\theta = 10$ as our unified, default setting, as it falls squarely within the stable flat region and consistently achieves robustly high detection performance across all test datasets without requiring dataset-specific tuning.

3) Analysis Under Different Noise Levels: In real-world applications, the HSI is often affected by noise during collection and transmission, which makes it difficult to detect targets accurately. To evaluate the robustness of different methods, we add Gaussian noise with variances of 0.001, 0.005, and 0.01 to the Airport-4 dataset. The testing results for all methods are listed in Table VII. As evident from the results, the proposed KBRTF consistently achieves the highest detection accuracy, maintaining AUC(P_D, P_F) values above 0.998 across all noise levels. In contrast, several competing methods, such as LSDM-MoG, exhibit significant performance degradation as noise intensity increases. This demonstrates that KBRTF is

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TABLE VII

AUC VALUES OF (P_D, P_F) OBTAINED BY DIFFERENT COMPARISON METHODS ON THE AIRPORT-4 DATASET UNDER DIFFERENT NOISE LEVELS

Noise levels	RX	LSMAD	LRASR	GTVLRR	PTA	PCA-TLRSR	CF2-GNBRL	GAED	LSDM-MoG	KBRTF
0.001	0.9596	0.9861	0.9812	0.9733	0.9948	0.9903	0.9969	0.9679	0.9372	0.9982
0.005	0.9739	0.9860	0.9806	0.9755	0.9930	0.9899	0.9968	0.9670	0.9043	0.9984
0.01	0.9756	0.9870	0.9715	0.9773	0.9945	0.9909	0.9968	0.9682	0.8513	0.9983

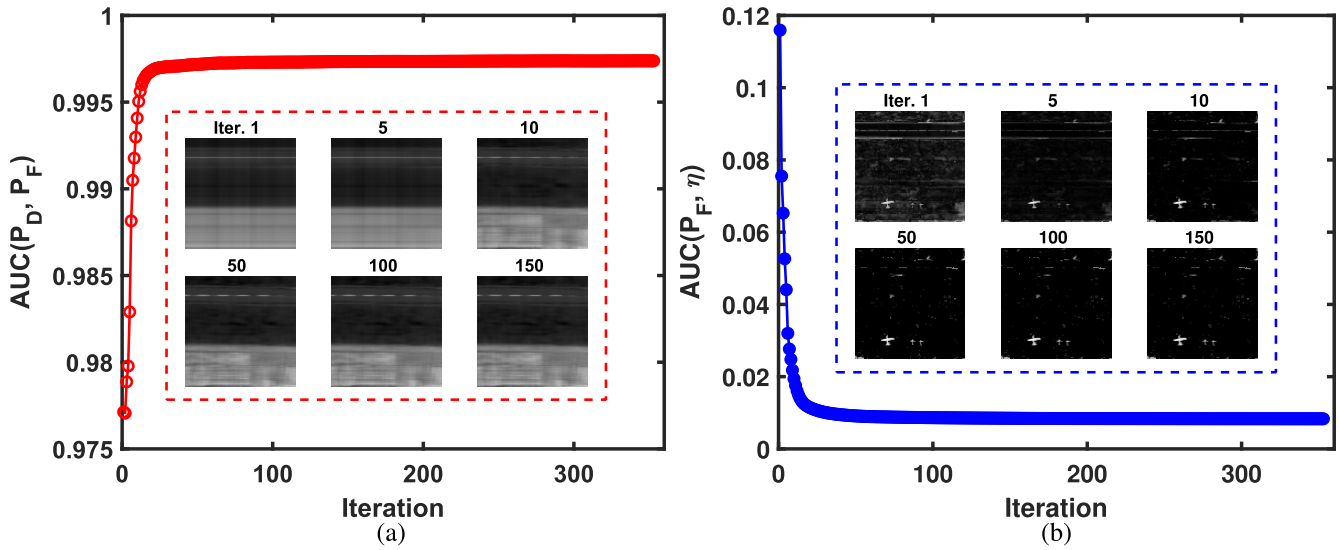


Fig. 13. Convergence of the proposed KBRTF method on the Airport-4 dataset. (a) Evolution of AUC(P_D, P_F) values and the corresponding background images. (b) Evolution of AUC(P_F, η) values and the resulting anomaly detection maps.

robust to noise and provides stable detection performance under noisy conditions.

4) Convergence: We evaluate the performance evolution of the proposed method on the Airport-4 dataset throughout the iterative process. To fully demonstrate the theoretical asymptotic convergence behavior, the maximum iteration limit was temporarily relaxed to 5000 for this specific test, and the algorithm automatically satisfied the termination criterion (25) and stopped at around 300 iterations. Fig. 13(a) illustrates the variation of the AUC(P_D, P_F) with respect to the number of iterations. The curve exhibits a steep increase at the initial stages, stabilizing around 64 iterations, and eventually converges to approximately 0.997. This observation theoretically validates our empirical decision to cap the maximum iteration at 100 in the standard experiments, as it perfectly captures the stable optimal performance while avoiding unnecessary computational overhead. In addition to the quantitative results, the corresponding background images at several iteration steps are presented in Fig. 13(a). Fig. 13(b) depicts the evolution of AUC(P_F, η) as the iterations proceed. Its value decreases sharply from an initial level higher than 0.1 and eventually stabilizes around 0.01. The associated anomaly detection maps visually support this trend. Overall, these results demonstrate that the proposed method achieves satisfactory convergence behavior and exhibits strong stability.

5) Computational Complexity: The computational complexity of the proposed KBRTF is dominated by the posterior updates of the latent and factor tensors in each iteration.

TABLE VIII

DETECTION PERFORMANCE, EFFECTIVE TUBAL RANK, AND RUNNING TIME (S) OF KBRTF ON DIFFERENT SPATIAL CROP SIZES FROM THE QINGPU-II DATASET

Metrics	Spatial sizes (pixels)					
	120 × 120	160 × 160	200 × 200	240 × 240	280 × 280	320 × 320
AUC (P_D, P_F)	0.9997	0.9998	0.9998	0.9991	0.9988	0.9988
Effective tubal rank	3	4	7	10	14	18
Running time (s)	13.12	29.77	67.18	130.37	263.10	529.31

Specifically, the computation cost of updating the latent tensors $\mathcal{G}$ and $\mathcal{H}$ is $\mathcal{O}(r(n_1^3 + n_2^3) + rn_3(n_1^2 + n_2^2))$. The update of factor tensors $\mathcal{A}$ and $\mathcal{B}$ takes $\mathcal{O}(r^3 n_3^3 + rn_1 n_2 n_3^2)$. Therefore, the overall complexity of our algorithm is $\mathcal{O}(r(n_1^3 + n_2^3) + r\,n_3(n_1^2 + n_2^2) + r^3 n_3^3 + r\,n_1 n_2 n_3^2)$. It is worth noting that, in practice, both the retained dimension n_3 and the effective tubal rank r are typically much smaller than the spatial dimensions due to PCA preprocessing and automatic rank updating, which substantially alleviates the practical computational burden.

To further validate the theoretical complexity analysis and investigate the scalability of the proposed method on larger scenes, we design an empirical scaling experiment based on real hyperspectral imagery. In this experiment, we utilize the Qingpu-II dataset [76], which was acquired over the Shanghai region of China using the airborne multimodular imaging spectrometer. The original image has a spatial dimension of 740 × 400 pixels with 250 spectral bands covering the wavelength range of 400–1000 nm. The spatial resolution of the images is 0.75 m, and the black rooftops in the scene

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are considered the targets to be detected. Specifically, we first crop the original image from the top-left corner to generate subdatasets with spatial sizes progressively increasing from 120×120 to 320×320 pixels. Each cropped subdataset is then processed using PCA to retain $n_3 = 20$ spectral bands. Since the spatial dimensions vary across subdatasets, we set the initial tubal rank adaptively as $r = \min(n_1, n_2)$ for each case. All other experimental parameters remain consistent with those in Section V-B. As reported in Table VIII, the running time naturally grows as the spatial size increases, which aligns with the $\mathcal{O}(n_1^3 + n_2^3)$ complexity of the spatial matrix inversions. However, as also shown in the table, the automatic rank determination mechanism effectively prunes the massive initial rank down to a very small rank, keeping the r-related computations highly efficient. Despite the increased computational burden, the proposed method demonstrates remarkable robustness, maintaining highly stable and excellent detection accuracy (AUC(P_D, P_F) around 0.999) across all tested spatial sizes. To further mitigate the computational bottleneck on extremely large-scale HSIs in future work, acceleration strategies such as patch-based processing and fast iterative solvers (e.g., preconditioned conjugate gradient) can be readily adopted to replace the exact matrix inversions.

6) Ablation Study: To thoroughly identify the individual contribution of each key design choice in KBRTF, we conduct a series of controlled ablation experiments across all eight HSI datasets. Specifically, we implement four different variants of KBRTF to evaluate the significance of our model components and kernel configurations.

1) *No-Kernel Variant:* We remove the spatial prior correlation by setting the kernel matrices to identity matrices, that is, $\mathbf{K_G} = \mathbf{I}_{n_1}$ and $\mathbf{K_H} = \mathbf{I}_{n_2}$, evaluating the importance of utilizing spatial prior knowledge.
2) *Element-Wise Sparsity Variant:* We replace the proposed tube-wise joint sparse prior on the anomaly tensor $\mathcal{S}$ with an element-wise (entry-wise) sparse prior, assuming each entry $\mathcal{S}_{ijk}$ is mutually independent

$$.p(\mathcal{S}|\boldsymbol{\xi}) = \prod_{i=1}^{n_1}\prod_{j=1}^{n_2}\prod_{k=1}^{n_3}\mathcal{N}\left(\mathcal{S}_{ijk} \mid 0,\ \xi_{ijk}^{-1}\right)$$
$$p(\boldsymbol{\xi}) = \prod_{i=1}^{n_1}\prod_{j=1}^{n_2}\prod_{k=1}^{n_3}\mathrm{Ga}\left(\xi_{ijk}\middle|a_0^{\xi}, b_0^{\xi}\right)$$

where ξ_{ijk} is the entry-wise precision parameter. Symmetrically, the corresponding variational posterior distributions $q(\mathcal{S}_{ijk}) = \mathcal{N}(\mathcal{S}_{ijk}|\mathbb{E}[\mathcal{S}_{ijk}], \Sigma^{\mathcal{S}_{ijk}})$ and $q(\xi_{ijk}) = \mathrm{Ga}(\xi_{ijk}|a^{\xi_{ijk}}, b^{\xi_{ijk}})$ are derived as

$$\begin{aligned}
\Sigma^{\mathcal{S}_{ijk}} &= \left(\mathbb{E}[\tau] + \mathbb{E}\left[\xi_{ijk}\right]\right)^{-1} \\
\mathbb{E}\left[\mathcal{S}_{ijk}\right] &= \Sigma^{\mathcal{S}_{ijk}}\mathbb{E}[\tau]\left(\mathcal{Y}_{ijk} - \mathbf{U}_{:k}^{H}\left(\mathbb{E}[\mathcal{A}_{i::}] \vartriangle \mathbb{E}\left[\mathcal{B}_{j::}^{H}\right]\right)\right) \\
a^{\xi_{ijk}} &= a_0^{\xi} + \frac{1}{2} \\
b^{\xi_{ijk}} &= b_0^{\xi} + \frac{1}{2}\left(\mathbb{E}\left[\mathcal{S}_{ijk}\right]^2 + \Sigma^{\mathcal{S}_{ijk}}\right).
\end{aligned}$$

During implementation, the initial precision tensor $\boldsymbol{\xi}$ is directly initialized as rand(n_1, n_2, n_3).

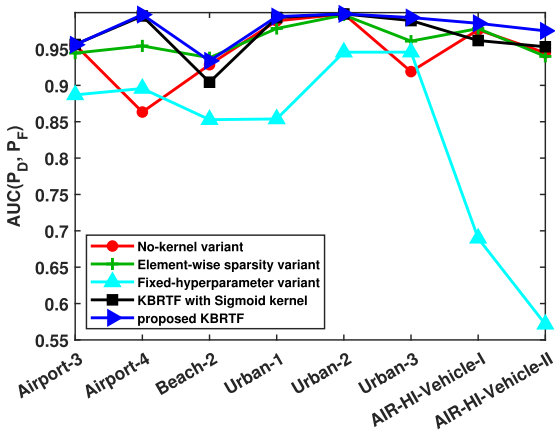


Fig. 14. Ablation study of proposed KBRTF under different variants on eight datasets.

3) *Fixed-Hyperparameter Variant:* We fix the key hyperparameters $\boldsymbol{\lambda}$, $\boldsymbol{\xi}$, τ, σ_g, and σ_h to pre-set constant values instead of updating them dynamically within the Bayesian framework.
4) *KBRTF With Sigmoid Kernel:* We employ the Sigmoid kernel formulated as $\mathbf{K}_{i,j} = \tanh(c \cdot i \cdot j + d)$ with $c = 0.0001$ and $d = 1$ to replace the proposed Gaussian kernel.

The results are presented in Fig. 14. It can be observed that all variants exhibit performance degradation on multiple datasets compared with the complete KBRTF model. In particular, removing the kernel structure generally reduces the detection performance, demonstrating the effectiveness of exploiting spatial correlations through spatial priors. Replacing the proposed tube-wise sparsity model with an element-wise sparsity formulation also leads to inferior results, indicating the importance of preserving the structural sparsity characteristics of anomalies. Furthermore, fixing the hyperparameters causes noticeable performance deterioration on several datasets, highlighting the advantage of Bayesian inference in adaptively learning model complexity from data. The comparison between the KBRTF with Sigmoid kernel and KBRTF further shows that the Gaussian kernel achieves the best overall performance, validating the adopted kernel design in the proposed framework.

VI. Conclusion

This article considers the low-rank background term, the tube-wise sparsity anomalies term, and the Gaussian noise term to develop a novel KBRTF method for HAD. Specifically, the low-rank structure is captured by decomposing the tensor into two smaller factor tensors, each influenced by a latent tensor and a Gaussian kernel. This joint modeling strategy effectively captures the spectral–spatial correlation and local smoothness characteristics of the hyperspectral background. The hierarchical tube-wise sparsity-inducing prior and Gaussian noise distribution can better extract anomalies from the background. Additionally, an efficient variational Bayesian algorithm is developed to perform model inference. Extensive experiments on eight hyperspectral datasets demonstrate that

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KBRTF achieves superior detection accuracy and background suppression capabilities.

Future research will focus on addressing several limitations of the current framework. First, the spatial bandwidth parameter θ is manually specified, and developing adaptive kernel learning strategies within the Bayesian framework remains an important direction for future work. Second, the variational inference procedure may still become computationally demanding for extremely large-scale HSIs. Therefore, incorporating acceleration strategies such as patch-based processing and efficient iterative solvers will be investigated to further improve scalability. In addition, extending the framework to more complex noise models and broader hyperspectral analysis tasks is another promising research direction.

APPENDIX

The appendix provides the detailed derivations for the posterior distributions of the latent variables and hyperparameters in the proposed KBRTF method, based on the general update rule in (14).

The variational posterior for the latent tensor $\mathcal{G}$ is obtained by taking the expectation of the log-joint probability with respect to all other variables in $\boldsymbol{\Psi} \setminus \mathcal{G}$. The terms dependent on $\mathcal{G}$ are the priors $p(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g)$ and $p(\mathcal{G}|\boldsymbol{\lambda})$. The log-posterior is

$$\begin{aligned}\ln q(\mathcal{G}) &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\mathcal{G}}\left[\ln p\left(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g\right) + \ln p\left(\mathcal{G}|\boldsymbol{\lambda}\right)\right] + \text{const} \\ &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\mathcal{G}}\left[\sum_{l=1}^{r}\sum_{k=1}^{n_3}\left(-\frac{\sigma_g}{2}\left\|\mathcal{A}_{:lk} - \mathbf{K_G}\mathcal{G}_{:lk}\right\|^2 \right.\right. \\ &\quad \left.\left. - \frac{\lambda_l}{2}\left\|\mathcal{G}_{:lk}\right\|^2\right)\right] + \text{const}.\end{aligned}$$

By isolating the terms for each column $\mathcal{G}_{:lk}$ and completing the square, we obtain a quadratic form in $\mathcal{G}_{:lk}$

$$\begin{aligned}\ln q(\mathcal{G}_{:lk}) = &-\frac{1}{2}\mathcal{G}_{:lk}^H\left(\mathbb{E}\left[\sigma_g\right]\mathbf{K_G}^H\mathbf{K_G} + \mathbb{E}\left[\lambda_l\right]\mathbf{I}_{n_1}\right)\mathcal{G}_{:lk} \\ &+ \mathcal{G}_{:lk}^H\left(\mathbb{E}\left[\sigma_g\right]\mathbf{K_G}^H\mathbb{E}\left[\mathcal{A}_{:lk}\right]\right) + \text{const}.\end{aligned}$$

This corresponds to the log-density of a Gaussian distribution. The derivation for $\mathcal{H}$ is analogous due to the symmetric structure of the model.

The posterior for $\mathcal{A}$ depends on the likelihood $p(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau)$ and its prior $p(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g)$. Focusing on the vectorized ith horizontal slice $\mathbf{a}_i = \text{vec}(\mathcal{A}_{i::})$, the log-posterior is

$$\begin{aligned}&\ln q(\mathbf{a}_i) \\ &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\mathbf{a}_i}\left[\ln p\left(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau\right) + \ln p\left(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g\right)\right] + \text{const} \\ &= -\frac{1}{2}\mathbf{a}_i^H\left(\mathbb{E}\left[\tau\right]\sum_{j=1}^{n_2}\mathbb{E}\left[\left(T_j^{\mathcal{B}}\right)^H T_j^{\mathcal{B}}\right] + \mathbb{E}\left[\sigma_g\right]\mathbf{I}_{rn_3}\right)\mathbf{a}_i \\ &\quad + \mathbf{a}_i^H\left(\mathbb{E}\left[\tau\right]\sum_{j=1}^{n_2}\mathbb{E}\left[T_j^{\mathcal{B}}\right]^H\left(\mathcal{Y}_{ij:} - \mathbb{E}\left[\mathcal{S}_{ij:}\right]\right)\right. \\ &\quad \left. + \mathbb{E}\left[\sigma_g\right](\mathbf{K_G})_{i:}\,\mathbb{E}\left[\mathcal{G}_{(1)}\right]\right) + \text{const}\end{aligned}$$

where $T_j^{\mathcal{B}} = (\mathbf{U}^H \triangle \mathcal{B}_{j::})_{(1)}$. This quadratic form indicates that $q(\mathbf{a}_i)$ is a Gaussian distribution. The posterior for $\mathbf{b}_j = \text{vec}(\mathcal{B}_{j::})$ is derived similarly.

The posterior for the anomaly tensor $\mathcal{S}$ depends on the likelihood $p(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau)$ and its own prior $p(\mathcal{S}|\boldsymbol{\xi})$. For each tube vector $\mathcal{S}_{ij:}$, the log-posterior is

$$\begin{aligned}\ln q\left(\mathcal{S}_{ij:}\right) &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\mathcal{S}_{ij:}}\left[\ln p\left(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau\right) + \ln p\left(\mathcal{S}|\boldsymbol{\xi}\right)\right] + \text{const} \\ &= -\frac{1}{2}\mathcal{S}_{ij:}^H\left(\mathbb{E}\left[\tau\right] + \mathbb{E}\left[\xi_{ij}\right]\right)\mathcal{S}_{ij:} + \mathcal{S}_{ij:}^H\mathbb{E}\left[\tau\right]\left(\mathcal{Y}_{ij:}\right. \\ &\quad \left. - \mathbf{U}^H\left(\mathbb{E}\left[\mathcal{A}_{i::}\right] \triangle \mathbb{E}\left[\mathcal{B}_{j::}\right]^H\right)\right) + \text{const}.\end{aligned}$$

This is the log-density of a Gaussian distribution for $\mathcal{S}_{ij:}$.

The posterior distributions of all precision parameters follow Gamma distributions, owing to the conjugate relationship between the Gamma and Gaussian distributions. The log-posterior for λ_l depends on $p(\mathcal{G}|\boldsymbol{\lambda})$, $p(\mathcal{H}|\boldsymbol{\lambda})$, and $p(\boldsymbol{\lambda})$

$$\begin{aligned}&\ln q(\lambda_l) \\ &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\lambda_l}\left[\ln p\left(\mathcal{G}|\boldsymbol{\lambda}\right) + \ln p\left(\mathcal{H}|\boldsymbol{\lambda}\right) + \ln p\left(\boldsymbol{\lambda}\right)\right] + \text{const} \\ &= \left(\frac{(n_1+n_2)n_3}{2} + a_0^{\lambda} - 1\right)\ln\lambda_l \\ &\quad - \lambda_l\left(\frac{1}{2}\sum_{k=1}^{n_3}\left(\mathbb{E}\left[\left\|\mathcal{G}_{:lk}\right\|^2\right] + \mathbb{E}\left[\left\|\mathcal{H}_{:lk}\right\|^2\right]\right) + b_0^{\lambda}\right) + \text{const}.\end{aligned}$$

This is the log-density of a Gamma distribution $\text{Ga}(\lambda_l|a^{\lambda_l}, b^{\lambda_l})$.

For anomaly precisions $\boldsymbol{\xi}$, the log-posterior for ξ_{ij} depends on $p(\mathcal{S}|\boldsymbol{\xi})$ and $p(\boldsymbol{\xi})$

$$\begin{aligned}\ln q\left(\xi_{ij}\right) &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\xi_{ij}}\left[\ln p\left(\mathcal{S}|\boldsymbol{\xi}\right) + \ln p\left(\boldsymbol{\xi}\right)\right] + \text{const} \\ &= \left(\frac{n_3}{2} + a_0^{\xi} - 1\right)\ln\xi_{ij} - \left(\frac{1}{2}\mathbb{E}\left[\left\|\mathcal{S}_{ij:}\right\|^2\right] + b_0^{\xi}\right)\xi_{ij} \\ &\quad + \text{const}.\end{aligned}$$

This corresponds to a Gamma distribution $\text{Ga}(\xi_{ij}|a^{\xi_{ij}}, b^{\xi_{ij}})$.

For kernel precisions σ_g and σ_h, the log-posterior for σ_g depends on $p(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g)$ and its Jeffreys prior $p(\sigma_g) \propto \sigma_g^{-1}$

$$\begin{aligned}\ln q\left(\sigma_g\right) &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\sigma_g}\left[\ln p\left(\mathcal{A}|\mathcal{G}, \mathbf{K_G}, \sigma_g\right) + \ln p\left(\sigma_g\right)\right] + \text{const} \\ &= \left(\frac{n_1 r n_3}{2} - 1\right)\ln\sigma_g - \frac{1}{2}\mathbb{E}\left[\left\|\mathcal{A} - \mathcal{G}\times_1\mathbf{K_G}\right\|_F^2\right]\sigma_g \\ &\quad + \text{const}.\end{aligned}$$

This is the log-density of a Gamma distribution. The derivation for σ_h is analogous.

For noise precision τ, the log-posterior for τ depends on the likelihood $p(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau)$ and its prior $p(\tau)$

$$\begin{aligned}&\ln q(\tau) \\ &= \mathbb{E}_{\boldsymbol{\Psi}\setminus\tau}\left[\ln p\left(\mathcal{Y}|\mathcal{A}, \mathcal{B}, \mathcal{S}, \tau\right) + \ln p\left(\tau\right)\right] + \text{const} \\ &= \left(\frac{n_1 n_2 n_3}{2} + a_0^{\tau} - 1\right)\ln\tau \\ &\quad - \left(\frac{1}{2}\sum_{i}^{n_1}\sum_{j}^{n_2}\mathbb{E}\left[\left\|\mathcal{Y}_{ij:} - \mathcal{S}_{ij:} - \mathbf{U}^H\left(\mathcal{A}_{i::} \triangle \mathcal{B}_{j::}^H\right)\right\|^2\right] + b_0^{\tau}\right)\tau \\ &\quad + \text{const}.\end{aligned}$$

This also corresponds to a Gamma distribution $\text{Ga}(\tau|a^{\tau}, b^{\tau})$.

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